



Recent advances in wayside condition monitoring for railways: a comprehensive review

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Abstract

The demand for faster, longer, and safer railway networks has intensified the need for robust condition monitoring systems, particularly those capable of detecting rolling stock defects such as wheel flats, bearing faults, hunting movement, overloading, and unbalanced loads. Effective monitoring and timely maintenance are essential to mitigate these issues, enhancing both reliability and safety in railway operations. This paper presents a comprehensive review of wayside condition monitoring (WCM) systems. The study begins with a structured overview of WCM architectures, followed by a bibliometric analysis that highlights recent research trends in machine learning applications for railway condition monitoring. A detailed classification of rolling stock defects is provided to establish the foundation for condition assessment. The review then presents an extensive survey of commercial wayside monitoring systems currently in use, including hot axle box detectors, wheel impact load detectors, acoustic detection systems, and weigh-in-motion technologies. The various sensor types integrated into these systems are also described in detail. To bridge the gap between raw data collection and actionable insights, the paper includes a dedicated section on knowledge extraction from WCM systems. This section outlines key approaches for fault detection, diagnosis, and classification using advanced data processing methods, including signal processing and machine learning techniques. Finally, the paper identifies critical challenges such as data quality, real-time processing constraints, infrastructure limitations, and the need for generalizable models. It also discusses research gaps and suggests future directions, including the integration of edge computing, digital twin technology, and self-diagnostic capabilities to support the development of next-generation intelligent WCM systems.

Keywords Artificial intelligence · Fault diagnosis · Railways · Wayside condition monitoring · Sensors

1 Introduction

Railway transportation contributes to economic growth and development, and it produces fewer carbon emissions than roads. The EU aims to be climate-neutral by 2050 [1], and the expansion of the railway transport sector can play an important role in the transition to a climate-neutral society. To ensure that rail transport runs as safely and smoothly

as possible, both trains and tracks should be monitored, inspected, and maintained to assess their condition and schedule maintenance or repair actions.

It is crucial to maintain wheel condition through advanced monitoring techniques to ensure vehicle safety and optimal performance. Condition monitoring allows for identifying damage, which is crucial to minimize the risk in railway transportation and maximize train availability. Generally, condition monitoring involves data collection, data analysis, alert generation, and maintenance planning [2–4]. The integration of internet of things (IoT) technology allows for real-time data collection, enabling continuous monitoring of wheel conditions [5]. Data collection comprises gathering information from sensors that will be analyzed for damage identification [6]. Different types of artificial intelligence techniques [7] may be used for wheel damage identification, particularly machine learning [8–10], including deep

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learning [11, 12], and image recognition [13]. The integration of machine learning (ML) techniques in WCM has revolutionized the field of predictive maintenance and asset management within the rolling stock industry [14]. It has been demonstrated that machine learning, particularly deep learning, exhibits improved performance and accuracy at fault diagnosis and predictive maintenance scheduling [15–17].

When abnormal situations are detected, i.e., when measurements exceed some predefined thresholds, specific alerts are prompted. Once damage is detected, diagnosis and classification can be performed to schedule effective maintenance plans. Beyond diagnosing failures for corrective maintenance [18], the condition monitoring system can also provide prognostics to predict future failures and estimate the remaining useful life of the wheels [2].

The vehicle condition assessment can be categorized into two types: offline inspection and online monitoring. Inspections are usually discrete events, while monitoring is an ongoing process. In inspection, the vehicle is placed out of operation to perform a maintenance check, while in monitoring, the vehicle is monitored in operation, providing insights into how the vehicle operates over time [19]. Different techniques are typically used in vehicle offline inspection, such as ultrasonic [20] and infrared cameras [21]. Online monitoring can be categorized into onboard [22] and trackside [23]. Onboard monitoring implies that sensors are mounted on the vehicle [24, 25], whereas trackside or wayside systems are placed on the track or close to it to monitor the condition of vehicle components while the train is in operation. This type of monitoring can use ultrasonic, acoustic, vibration techniques, strain gauges, fiber optic sensing, lasers and thermal and high-speed cameras [7].

Currently, wayside systems are the preferred solution to detect damage since all vehicles can be monitored during their passage in a specific track location [17, 26–29]. These systems can measure wheel impacts on the rail surface, monitor bearing faults, and measure the relative position of the wheelset to the rail to identify inter-axle misalignment, hunting movement, angle of attack and lateral shift [9, 30, 31]. A review of existing condition monitoring techniques, up to 2012, dedicated to monitoring railway vehicle dynamics can be found in Refs. [32, 33].

Many types of vehicle damage can affect train operational performance, with defective wheels being among the most critical issues. Wheel defects can damage the track and increase the risk of derailments. They may occur on the wheel tread, manifesting as flats, spalling, or shelling, or along the wheel perimeter, leading to issues such as wheel corrugation and polygonal wear [27]. Comprehensive reviews of wheel defects are provided by Asplund et al. [34] and in Engineering Standard Rolling Stock [35, 36]. Similarly, bearing faults can be detected using hot axle box detectors, as well as through analysis of acoustic emission

and vibration signals [37, 38]. A review of the detection of bearing faults can be found in Refs. [39–42]. Furthermore, the presence of hunting movement not only reduces vehicle ride quality but also can increase the risk of wheelset derailment. Recent studies have introduced methods for detecting hunting using wayside measurements [43, 44] or onboard signals [45–47].

The synthesis of monitoring technologies and ML techniques presented in this review provides a roadmap for addressing the limitations of current WCM systems, setting the stage for future innovations in railway safety and efficiency. This paper comprehensively reviews wayside condition monitoring systems, emphasizing the integration of ML techniques for enhanced defect detection and maintenance planning. The paper identifies trends and research gaps in ML applications for railway condition monitoring through bibliometric analysis. Key areas explored include commercial WCM solutions, common defects, and ML-based software frameworks. Finally, the study discusses the challenges, opportunities, and future directions for advancing WCM technologies to meet the growing demands of modern railways.

Hence, this research aims to address the following questions:

- What are the actual commercial wayside systems for vehicle condition diagnosis? What types of sensors are used, and how do they function?
- What are the most recent artificial intelligence techniques used for identifying wheel damage?
- What bibliographic trends exist in relation to wayside condition monitoring?
- What are the existing challenges, gaps, and opportunities for future innovations in wayside condition monitoring?

2 Bibliometric performance analysis

This section presents a bibliometric analysis of machine learning applications in railway condition monitoring, based on 471 Scopus records retrieved from the Scopus database using the keywords “railways”, “condition monitoring”, and “machine learning”. Figure 1 provides an overview of publication trends, document types, and journal contributions. Figure 1a shows a sharp rise in annual publications from 1993 to 2024, reflecting growing interest in ML for railway monitoring. As shown in Fig. 1b, research articles dominate the literature (67%), followed by review articles (21%). Figure 1c highlights the most prominent journals in this field, such as *IFAC-PapersOnLine*, *Engineering Applications of Artificial Intelligence*, *Reliability Engineering & System Safety*, *Measurement*, and *Mechanical Systems and*

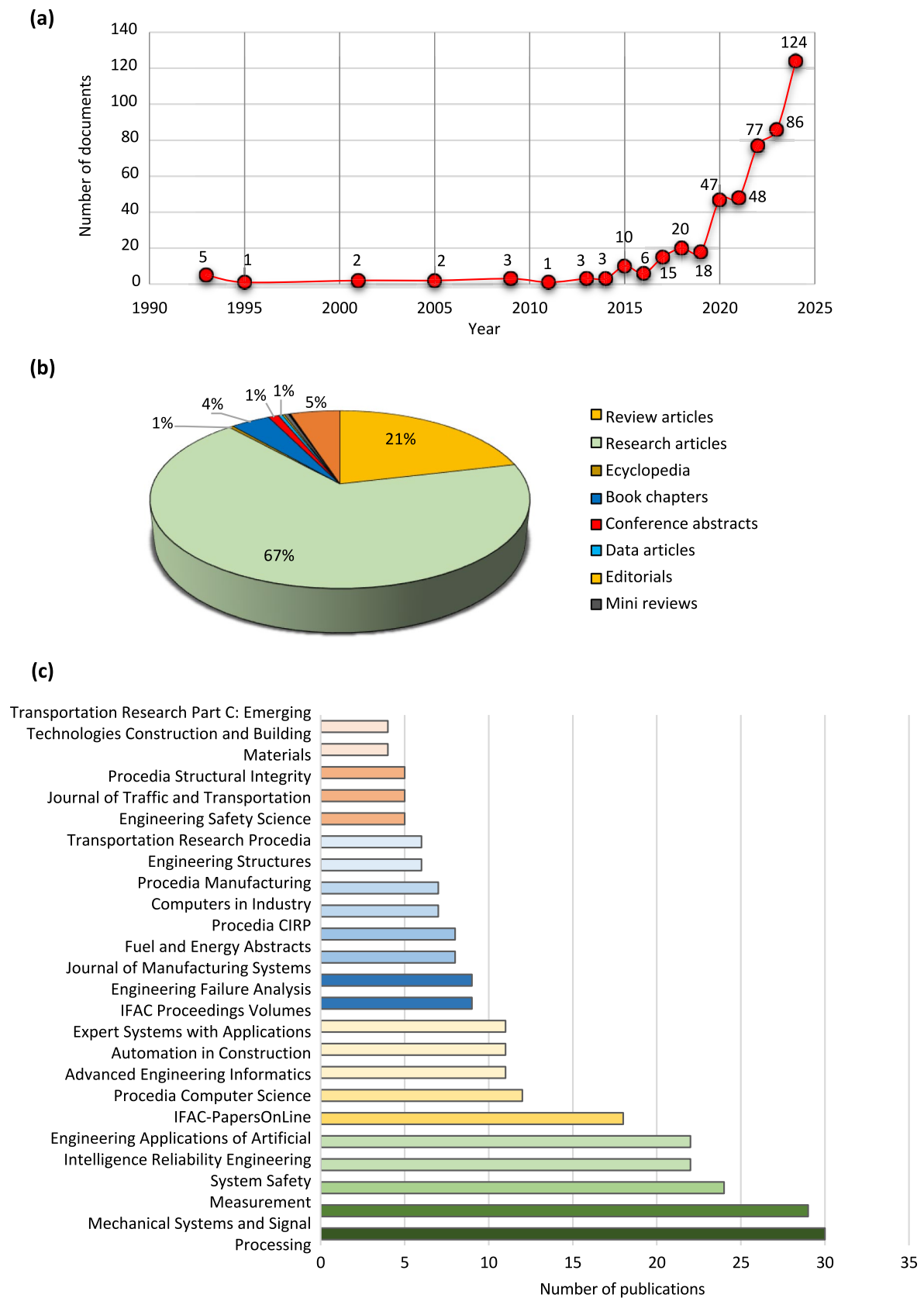


Fig. 1 Overview of ML research across condition monitoring and railways: **a** number of documents per year; **b** percentage of publications per document type; **c** number of publications per journal

Signal Processing, each contributing approximately 30 relevant publications.

Overall, the bibliometric analysis demonstrates the rapid expansion of ML applications in railway engineering, particularly in condition monitoring. The growing interest in this field is further evidenced by the nine additional articles already scheduled for publication in journal editions for 2025, identified through the keyword-based research. The findings underscore the need to further explore and integrate advanced techniques in this domain.

Keywords act as essential linguistic elements linking diverse publications and forming the central theme of this review paper. A total of 1,623 keywords associated with “railways”, “condition monitoring”, and “machine learning” were identified. It is worth noting that some of these keywords appeared in multiple forms (e.g., “*Internet of Things*” and “*IoT*”), which were not standardized, as shown in Fig. 2. For this review, selected keywords were analyzed

using VOSviewer software. From the initial set of 1623 keywords, only those with a minimum occurrence threshold of 3 were considered, resulting in 106 refined keywords. The relationships and connections among these keywords are visually represented in the cluster diagram of Fig. 2.

Cluster analysis reveals that “*Machine Learning*” dominates the focus, represented by a large red cluster. In addition, although not visible in the image due to its graphic density, the other red cluster represents the keyword “*predictive maintenance*”, while the blue one corresponds to “*condition monitoring*”. These three clusters are emphasized by their larger size, reflecting their high frequency. Recent topics such as “*Industry 4.0*”, “*Digital Twin*”, and “*Internet of Things*” are also prominent, as evidenced by recent scientific studies within the field. This type of cluster analysis provides valuable insights into the focus of research in railway condition monitoring using machine learning. This growing research interest in machine learning applications

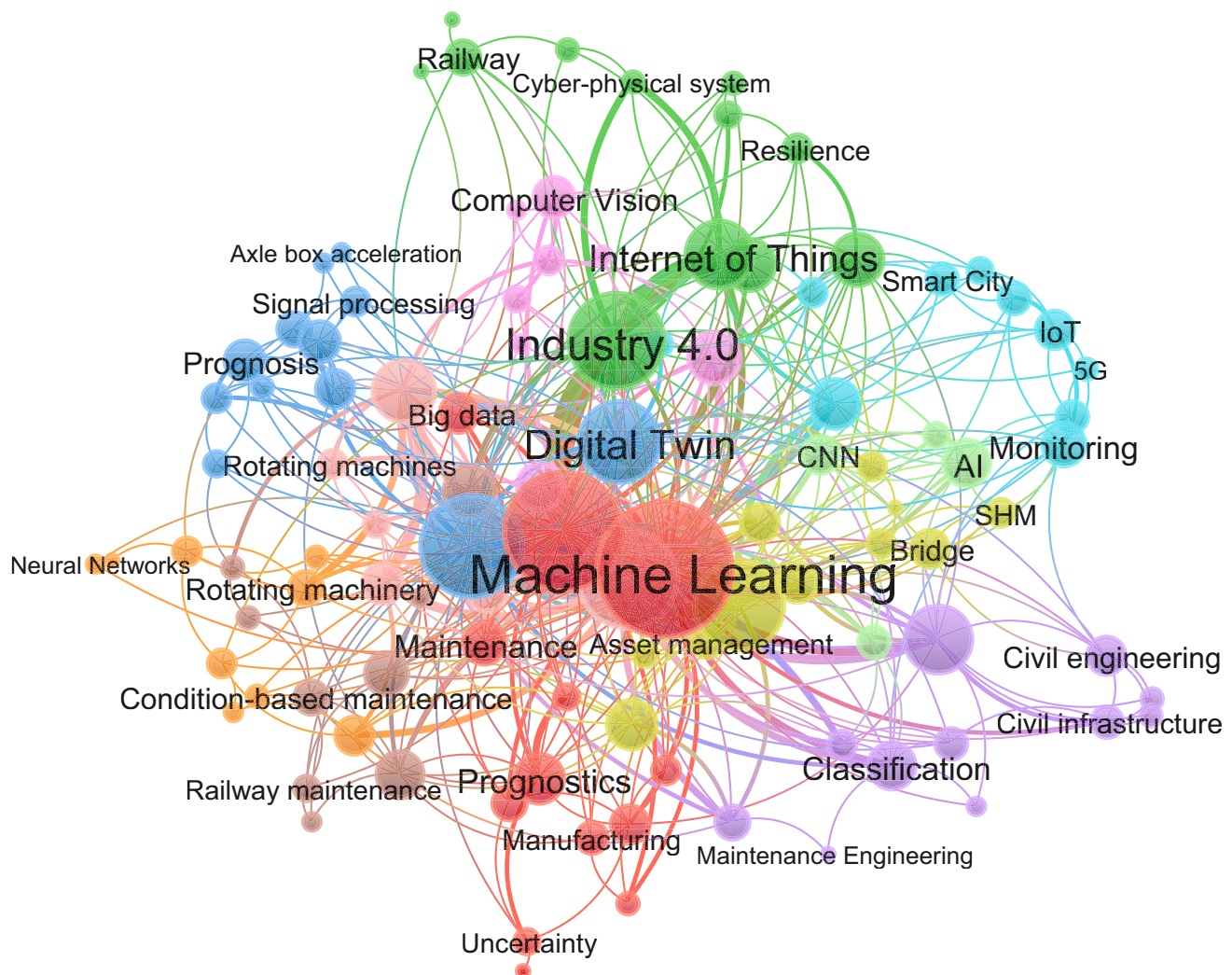


Fig. 2 Cluster diagram of the bibliometric analysis using VOSviewer software (version 1.6.20)

underscores the need for exploring their evolution, from foundational algorithms to modern neural networks, to address the complex requirements of WCM systems. The next section explores the historical development of machine learning, highlighting its progression and relevance to railway monitoring.

3 The evolution of machine learning techniques

Machine learning, a subset of artificial intelligence (AI), has evolved dramatically, finding applications in complex fields like railway systems. The roots of AI date back to the 1950s when Alan Turing introduced the Turing test, a measure to assess a machine's ability to exhibit intelligent behavior comparable to that of a human. The term artificial intelligence was later formalized by John McCarthy in 1956 to describe efforts to replicate human-like cognition in machines [48]. In 1957, Rosenblatt [49] developed the perceptron, an early neural network model used for tasks like speech recognition [50], cost estimation [51] and classification problems [52]. However, its capabilities were limited by the technology of the time. The 1960s and 1970s saw the development of foundational algorithms like the k -nearest neighbor algorithm [53] and the backpropagation algorithm [54], highlighting machine learning's growing potential.

The term machine learning became widely recognized in the 1980s with the introduction of the multi-layer perceptron (MLP) [55]. ML enables computers to learn and improve from data using methods like supervised, unsupervised, and reinforcement learning (RL), which are detailed later in this section [56]. The rise of deep learning (DL) in the 2000s, particularly around 2006, introduced models capable of recognizing complex patterns and making predictions independently [55]. These models utilized deep neural networks (DNNs) for tasks such as classification, regression, and image recognition [57]. Figure 3 illustrates a chronological representation of AI growth, highlighting the principal markers.

Machine learning algorithms have evolved through various self-learning methods, such as supervised, unsupervised, and reinforcement learning. Supervised learning trains models on labeled data to map inputs to outputs, commonly used in tasks like prediction. It includes regression and classification algorithms. Regression methods include linear regression and polynomial regression, while classification methods comprise k -nearest neighbor (k -NN), support vector machines (SVM), and decision trees (DT). The KNN algorithm, introduced in 1967, classifies data based on its nearest neighbors using distance metrics [50, 58]. SVM, developed in the 1990s, separates classes by defining a decision boundary and maximizing the margin for better generalization [59, 60]. Decision trees,

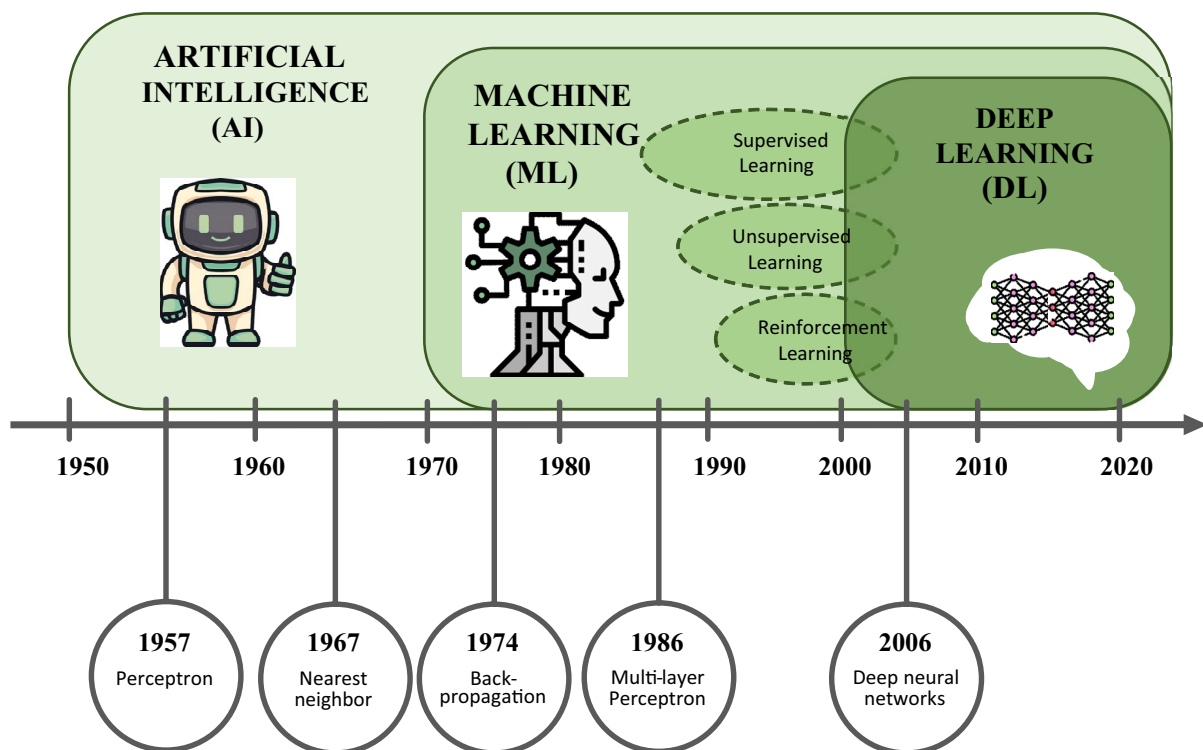


Fig. 3 Artificial intelligence development timeline

created in 1986 [61], are still popular for classification and decision-making [62].

Unsupervised learning focuses on discovering hidden patterns in data without using labeled information. Its main goals are clustering and dimensionality reduction, which do not require specific training. The concept of clustering was introduced in 1954 in a study on anthropological data. Since then, several studies have been published to evaluate and refine clustering algorithms, as stated in Ref. [63]. Some categories of clustering techniques, such as hierarchical, partitional, grid-based, density-based, and soft clustering, are characterized as the most relevant [64, 65]. Among these, partitional clustering, particularly the k -means technique, is widely recognized for its effectiveness in classification problems [66]. Moreover, density-based clustering is particularly effective for identifying non-linear structural patterns in data, with the density-based clustering non-parametric algorithm (DBSCAN) being one of the key algorithms in this category. This method does not require a predefined number of clusters [67]. Principal component analysis (PCA) [68] is popular for simplifying complex data while preserving key patterns for dimensionality reduction.

Reinforcement learning trains models by interacting with an environment, using feedback in the form of rewards or penalties to refine decision-making [55]. The main advantage of these algorithms is their ability to detect novelties and identify new or unknown data not experienced during

the training. RL operates by selecting actions to maximize cumulative rewards based on trial-and-error, enabling adaptive decisions. The term “reinforcement learning” was borrowed from animal learning literature by Minsky in the 1960s [69]. Key RL algorithms include q-learning, SARSA, dyna-q, and temporal difference [70].

Driven by advancements in computing power and the limitations of traditional ML algorithms, researchers intensify into more complex neural network (NN) architectures. This led to the rise of variants such as spiking neural networks (SNNs), critic-based neural networks, and deep neural networks (DNNs) [55]. DNNs can learn key features from different data types and address complex problems due to the deep knowledge acquired by the high number of hidden layers and neurons per layer. Among the various types of DNNs, convolutional neural networks (CNNs) are especially known for their use in classification, regression, and image recognition [57]. The evolution of machine learning has progressed from early models to advanced deep learning techniques. Despite these advancements, applying machine learning to real-world WCM systems still presents challenges in bridging theory with practical needs. Figure 4 illustrates the comprehensive mind map, where the ML characterization is represented with its associated techniques.

Machine learning techniques, particularly supervised and unsupervised algorithms, have become crucial in the railway sector. They support domains such as maintenance and

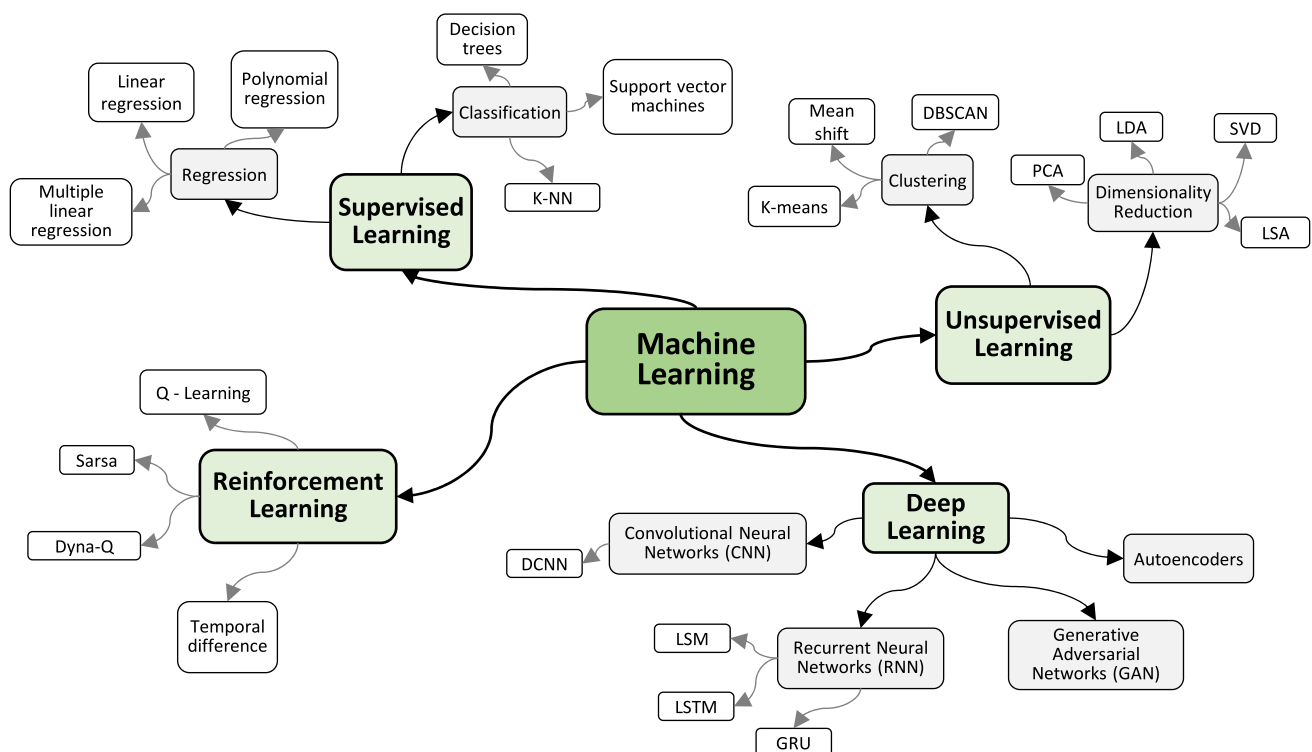


Fig. 4 Machine learning algorithms mind map

inspection, traffic planning, safety, autonomous control, and passenger mobility [71]. Some examples of these algorithms are SVM, PCA, clustering techniques, and NN. SVM has been used for detecting geometric defects [72], while ANN is used to predict track quality [73]. PCA is used for tasks such as evaluating track quality [74], maintenance analytics [75], and damage detection [27, 76–81]. Clustering-based methodologies have been applied to optimal maintenance planning for railway infrastructures [82], structural state analysis of railway bridges [83], and damage classification [8, 9, 30, 78, 84]. ANN was initially used for traffic event response and intelligent navigation management [85], pattern recognition for classifying rail flaws using ultrasonic images [86], predicting the safety of railway track sections [87], and correlating track structural conditions with data obtained from automated inspections [88].

Recently, several other applications have emerged in the railway sector. These applications primarily support condition monitoring of different railway components, such as rails [89], catenary systems [90, 91], track [90–93], and fasteners [94]. Deep neural networks have recently been applied for railway track region segmentation [95, 96] and obstacle detection [97, 98]. Additional insights into these advancements can be found in recent literature reviews, including studies by Phusakulkajorn et al. [99] and Tang et al. [71].

4 Rolling stock defects

Understanding the specific types of rolling stock defects lays the groundwork for developing targeted WCM solutions, enabling precise diagnosis and predictive maintenance. Defect detection using wayside monitoring in railways involves strategically placing monitoring equipment along the tracks to continuously assess the condition of

passing trains. Some common defects detected include wheel defects, bearing failures, brake malfunctions, and other critical issues such as hunting, overloading, and unbalanced loads, all of which pose significant risks to railway operations. Common wheel defects include flats, spalling, and shelling—typically localized on the wheel tread—as well as wheel corrugation and polygonal wheels, which manifest around the wheel perimeter (Fig. 5). Wheel defects cause increased friction and vibrations that negatively impact the performance and reliability of the train and track [100] and may lead to significant failures if not addressed at an early stage [101].

One of the most prevalent discrete defects on train wheels is flats (Fig. 5a), which occur when the wheel slides along the rail instead of rolling, causing increased friction, vibrations, and potential derailments [102–104]. Polygonal wheels (Fig. 5b), defined as a periodic irregularity around the wheel circumference [105], generate elevated dynamic vertical forces between the wheel and rail at specific frequencies of disturbance, which are influenced by the train's speed and the irregular wavelengths [106]. Shelling involves the detachment of material from the wheel tread due to thermal stresses or fatigue. Such damage can initiate latent cracks, that propagate unpredictably, potentially leading to sudden wheel failures [107, 108]. These defects collectively compromise wheel integrity and pose significant safety hazards to passengers and staff.

Defects in railway vehicle bearings (Fig. 6) can severely affect safety and operational efficiency. The bearings in railway vehicles can be damaged under various conditions, including improper lubrication, contamination, inadequate mounting, fatigue, and overheating [113]. These defects compromise the smooth rotation of the wheel, resulting in increased friction, noise, and vibration. If unaddressed, bearing defects can escalate into more severe issues such

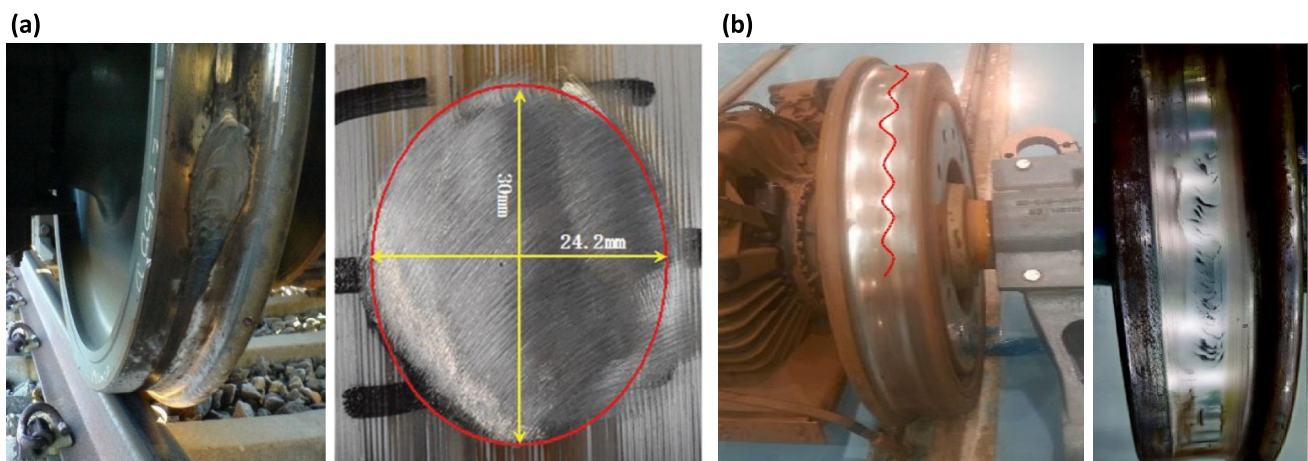


Fig. 5 Common wheel defects: **a** wheel flat [109, 110]; **b** polygonal wheel [111, 112]



Fig. 6 Defective bearing: **a** fractures of the inner ring of inner bearing; **b** damaged inner bearing; **c** fracture of the inner ring of outer bearing [114]

as overheating, axle failure, and potential derailment, posing significant safety risks and leading to costly repairs and operational downtime.

Hunting refers to the lateral movement of the wheelset on the rails, as shown in Fig. 7a, which can evolve to instability if the vehicle exceeds the critical speed. It manifests as a periodic motion involving the coupling of lateral displacement and yawing of the wheelset with increasing amplitudes [44]. Factors like wheelset design, wheel–rail wear, train speed, and weight distribution influence hunting instability. This phenomenon accelerates wear on wheels and rails, reduces passenger comfort due to vibrations, and increases safety risks, including derailments [115].

Overloading, unbalanced loads, and dynamic loads are critical factors that significantly affect the performance, safety, and longevity of railway vehicles. Overloading refers to a condition in which the total weight of the train exceeds its maximum design capacity, resulting from excessive cargo, misjudgement of load capacity, or economic pressures. Exceeding the train's weight limit can cause excessive wear on tracks, train wheels, and other components, leading to failures or accidents [119]. Overloaded trains can lead to accelerated degradation of infrastructure, such as tracks, bridges, and signaling systems, requiring more frequent maintenance or even early replacement.

An unbalanced load (Fig. 8) occurs when cargo weight is unevenly distributed across a railway vehicle. This imbalance, which can be longitudinal, transverse, or both, results from improper loading practices, shifts in cargo during transit, or variations in cargo weight. Primary causes include inadequate planning during loading, cargo movement due to vibrations, and vehicle design limitations. The effects of unbalanced loads are considerable, leading to accelerated wear on wheelsets and tracks, handling and stability issues that increase derailments risk, and compromised braking efficiency due to uneven weight distribution [81, 120].

Dynamic loads are caused by the forces exerted on railway vehicles during operation, arising from acceleration, deceleration, and track irregularities. Unlike static loads,

which remain constant, dynamic loads vary in magnitude and direction, affecting how weight is distributed during movement. These loads can arise from factors such as the train's speed, track conditions, and wheel–rail interactions. High dynamic loads can exacerbate the effects of unbalanced loads and overloading, leading to amplified stresses on the track and vehicle structure and an increased probability of derailments [121].

5 Wayside condition monitoring systems

5.1 Commercial systems

Several wayside condition monitoring systems in the rail industry are being employed to identify rolling stock defects and monitor vehicle operation. These systems can measure a wide range of physical parameters such as wheel temperature, axle load, wheel condition, vibration, noise, and other mechanical aspects. Automated inspection systems typically encompass the following devices: hot axle bearing and hot/cold wheel detectors, weigh-in-motion, wheel impact detectors, hunting detectors, acoustic bearing defect detectors, wheel profile detectors, and wheel tread condition monitoring detectors. An overview of the most common types of commercial wayside systems is presented in Table 1.

5.1.1 Hot axle box detector (HABD)

HABD is a common wayside detector system to monitor and identify hot bearings (due to bearing defects) using an infrared thermal detection system as wheelsets pass over the detector [124, 125]. An alarm is triggered if the difference between the infrared temperature of the bearing surface and the ambient temperature exceeds a predetermined threshold. However, these systems can only detect a faulty bearing when the fault is at an advanced stage [126]. It is common for rail companies to combine the HABD with a hot wheel detector (HWD) system to identify hot axles and wheels



Fig. 7 Wheel set hunting: **a** self-excited oscillation [116, 117]; **b** monitoring setup [118]

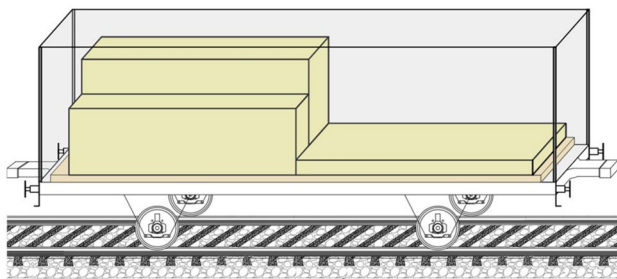


Fig. 8 Unbalanced load distribution on a freight wagon [81, 122, 123]

due to non-released hand brakes, incompletely released automatic brakes, and sticking brake shoes or disks. HABD and HWD use similar infrared scanning technology, though their decision algorithms and threshold criteria are different for detecting faulty bearings and seized brakes. Figure 9a shows an example of an HABD/HWD detector, where the hot box detector is installed at the axle box level to check temperature, and the hot wheel detector is installed at the track level to measure wheel temperature.

5.1.2 Trackside acoustic detector system (TADS)

TADS is based on an analysis of the acoustic signature generated by various components of a bearing, namely, inner ring (cone), outer ring (cup), and rollers. By analyzing these

Table 1 Most common types of commercial wayside systems

Type of System	Inspection features	Company brands
Hot axle-box detector (HABD) / hot wheel detector (HWD)	Aimed to detect damaged axle bearings (hot bearings) and seized brakes (hot wheels)	PHOENIX [132]; ITSS-AGUILA [133]; MERMEC [134]; TRACK-IQ [135]; PSI-TECHNICS [136]; SMARTSCAN-NG [137]; APNA-TECH [128]
Trackside acoustic detection system (TADS)	Monitoring the audible noise produced by the wheels and axle bearings of passing vehicles for defect identification	TADS-TM [129]; WABTEC- RailBAM [138]; TTCl [139]; APNA-TECH [128]; TRACK-IQ [135]
Wheel impact load detector (WILD)	Measuring the impact load of wheels using deformation sensors	ITSS-AGUILA [133]; LB-FOSTER [130]; WABTEC-WCM [138]; HITACHI-STC [140]; MERMEC [134]; ARGOS [141]; APNA-TECH [128]; TRACK-IQ [135]
Weight-in-motion (WIM) detector	Designed to capture vehicle dynamic weights in real-time and during motion	WABTEC-WCM [138]; HITACHI-STC [140]; ITSS-AGUILA [133]; MERMEC [134]; LB-FOSTER [130]; ARGOS [141]
Truck hunting detector (THD)	Identifying hunting movement on passing vehicles	TBOGI-HD [127]; ARGOS [141]; ARTC [142]; NAGORY-FOSTER [143]; TRACK-IQ [135]; PSI-TECHNICS [136]
Wheel profile measurement System (WPMS)	Controlling the exact profile of the wheel tread by using lasers or cameras	SelectraVision [131]; SMARTSCAN-NG [137]; WABTEC-WHEELVIEW [138]; TRIMBLE-TREADVIEW [144]; PSI-TECHNICS [136]

sounds, the system can precisely identify which bearing component is faulty. Acoustic arrays can improve measurement reliability by potentially being able to detect axle-bearing faults at earlier stages. The TADS employ arrays of microphones to monitor the audible noise produced by the axle bearings of passing rolling stock. The operating frequency of the microphones is typically 22–44 kHz, which means they are prone to errors arising from background noise as well as the measured train itself (e.g., wheel flats or engine noise). There are several commercial products available such as the TADS-TM system (Fig. 9b), which uses wayside microphones to detect high-risk defects in bearings and alert the driver as the train passes by the system.

5.1.3 Impact load detectors (WILD)

The most common products for monitoring the condition of train wheelsets are referred to as impact load detectors (WILD systems). The system records variations in wheel forces caused by the presence of irregularities in the wheel's shape (roundness deviations in the circumference). The typical types of deviation are classified and quantified (i.e., wheel flats, polygonization, shelling and welded spots). The WILD systems typically use strain gauge sensors or load cells applied to the rail, which accurately measure wheel forces. Despite the capabilities of current commercial systems in detecting and identifying wheel defects, they are still unable to diagnose specific types of defects and classify their severity at an early stage. Figure 9c shows an example of a WILD system consisting of strain gauges with multiple instrumented rail spans (between sleepers).

5.1.4 Weigh-in-motion (WIM)

Most wayside monitoring systems for detecting overloading or unbalanced loads rely on WIM techniques, which cannot determine the direction of the unbalanced situation (longitudinal or transverse) or assess its severity. Basically, the system continuously monitors the vertical forces the passing train applies to the rail. Unlike static scales, WIM systems are able to measure vehicles traveling at a reduced or normal traffic speed and do not require the vehicle to stop. The WIM system can be combined with a WILD system or installed as a stand-alone detector, as they share the same type of sensors (strain gauges or load cells).

5.1.5 Truck hunting detectors (THD)

THD is a wayside detector to evaluate a bogie's ability to maintain the centerline of the rail at relatively high speeds when operating empty or lightly loaded. There are two types of THD available in the market. The first system is based on rail strain gauges that measure lateral wheel forces on

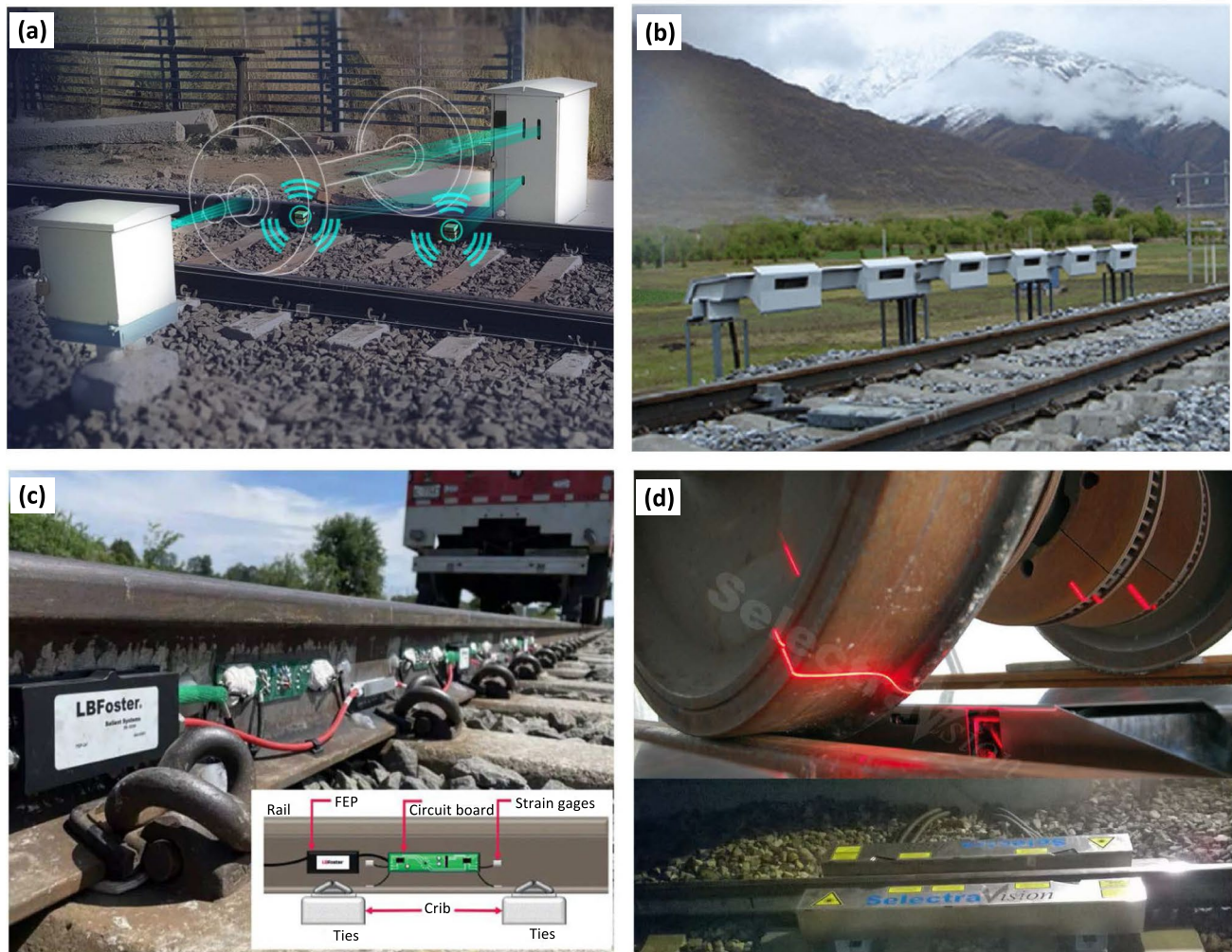


Fig. 9 Examples of commercial wayside systems: **a** hot box/hot wheel detector by APNA Technologies [128]; **b** TADS-TM system by Etion Digitise [129]; **c** WILD system by LB-Foster [130]; **d** wheel profile measuring system by SelectraVision [131]

a tangent track, while the other system is based on laser measurements of the angle of attack and the lateral position of an axle [127]. The force-based THD systems consist of a series of strain gages, which can be co-placed with a WILD system. On the other hand, the laser-based THD system consists of different modules. Each module contains an emitter and receiver for sending and receiving a laser beam. The THD thresholds are typically evaluated based on a hunting index (HI).

5.1.6 Wheel profile measurement systems (WPMS)

WPMS are based on scanning lasers or high-speed digital cameras to control the exact profile of the wheel tread. This system is typically used to measure wheel wear, flange height and thickness, rim thickness, and hollow tread issues. The data obtained can then be compared against a standard profile without wear or defect. The use of WPMSs eliminates

the need for manual profile inspection. Figure 9d shows an example of a WPMS that uses lasers and high-speed cameras to check the profile of the train's wheels with an accuracy of 0.5 mm.

5.2 Sensors

Wayside systems use a variety of non-destructive sensors to indirectly estimate wheel condition, loading condition and train operation status by measuring track responses. Common sensors include strain gauges, fiber optic sensors, accelerometers, ultrasonic sensors, acoustic sensors, as well as lasers and cameras.

5.2.1 Deformation sensors

The most commonly used type of deformation sensor is the strain gauge, which is normally used in WILD and WIM

systems. Strain gauges are configured to amplify very low measured strains (increase the signal-to-noise ratio) and are usually welded to the rail web near the neutral axis (see Fig. 10). Strain gauges offer unique advantages, providing precise local measurements and valuable data on how the tracks deform under various loads and conditions. However, measurements over longer lengths of track to understand the variability of a certain track parameter are less common. Furthermore, they are unsuitable for long-term monitoring as they are vulnerable to electromagnetic interference from environmental conditions.

Numerous researchers have investigated the application of strain gauges to detect wheel defects, estimate vehicle loads, and identify unbalanced loads in WIM systems. Stratman et al. [145] used a WILD system made up of 128 welded strain gauges, 32 strain gauges for vertical force measurements, and 32 strain gauges for lateral force measurements, per rail. Krummenacher et al. [146] used wheel load checkpoints with four strain gauges installed on the rail web for detecting and classifying wheel flats.

Another type of deformation sensors is the optical fiber sensors which can fall into two categories: fiber Bragg grating (FBG) sensors and distributed acoustic sensing (DAS) systems. FBG sensors contain gratings embedded along the fiber that reflect specific wavelengths of light when strained, enabling precise, point-based strain measurements. In contrast, DAS technology turns the entire optical fiber into a continuous sensor by interpreting Rayleigh and Brillouin backscattering patterns along the fiber length. This distinction makes DAS well-suited for distributed monitoring over large distances, whereas FBG is more targeted and localized.

Unlike the conventional strain gauges, FBG sensors (Fig. 11) offer immunity to electromagnetic interference, distributed sensing capabilities along the length of the rail, and a longer operational life. Using FBGs, a single optical fiber cable can connect an array of FBG strain sensors and transmit all the recorded data with minimal noise and

loss over long distances [147]. Liu et al. [148] developed a method for trackside monitoring of wheel conditions based on FBG. On the rail flange, two FBG arrays (one on each rail) were designed to capture the features of potential wheel defects. The online detector collects rail response data at a sampling rate of 5 kHz. The high sampling rate makes the sampling interval much shorter than the expected duration of localized anomalies, thereby ensuring that the desired features can be captured. In a study by Filograno et al. [149], field tests were performed on application of FBG sensors for monitoring railway traffic. A total of 20 FBG sensors were installed, 10 per rail. The system allows for dynamic measurement of the Bragg wavelength at sampling frequencies beyond 1 kHz, which is desirable in high-speed railways. Mishra et al. [150] proposed real-time monitoring and detection of the train wheel passage using an FBG sensor system. The system consists of four sensors installed vertically and eight sensors installed longitudinally at the rail flange (Fig. 11), with a sampling frequency of 2.5 kHz to interrogate all the FBGs simultaneously.

In contrast, DAS technology [151] is a relatively promising alternative for wayside condition monitoring, in which optical fiber serves as the sole medium for both information perception and transmission (Fig. 12). This technology is mainly based on the demodulation of Brillouin and Rayleigh scattering [151]. DAS enables real-time, long-range monitoring without the need for discrete physical sensors or additional power supplies along the track. Its immunity to electromagnetic interference, compatibility with existing telecom infrastructure, and scalability make it particularly attractive for monitoring high-traffic or remote railway corridors. Applications include train detection, axle positioning, and rail defect detection. Recent field studies underscore its practical effectiveness.

In a study by Gao et al. [152], DAS was applied to the quantitative rail strain measurement based on a series of field tests conducted in the Beijing Metro Line. The optical fiber

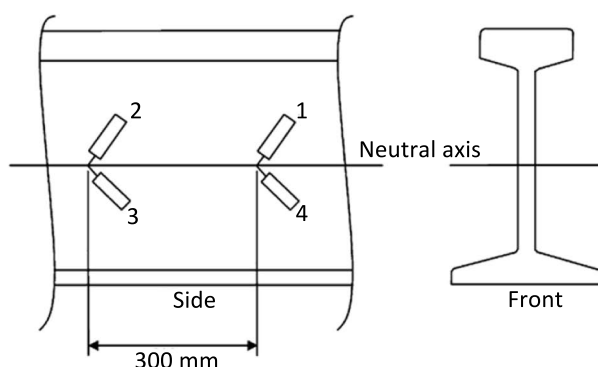


Fig. 10 Strain gauge setup used in the study of Stratman et al. [145]

Fig. 11 Deformation measurements performed in the rail using FBG sensors [150]



was laid along the rail at a height of 20 mm below the neutral axis, as illustrated in Fig. 12, and had a total length of 50 m. Other successful use cases include railway traffic monitoring [153, 154] and continuous vertical force assessment.

5.2.2 Vibration sensors

Accelerometers provide continuous, real-time data that can help identify wheel defects such as flats and out-of-round wheels. The most commonly used accelerometers are the highly sensitive piezoelectric types (Fig. 13). The raw acceleration data typically require pre-processing to filter noise and extract meaningful information.

Several authors have proposed algorithms for detecting wheel defects from acceleration measurements. Belotti et al. [156] proposed a diagnostic tool based on experimental dynamic rail responses obtained by four uniaxial vertical accelerometers and an inductive axle-counter block. Data from trains running at speeds between 10 and 100 km/h were acquired at a sampling frequency of 12.8 kHz and digitally converted by a 12-bit acquisition board. Salehi et al. [157] studied the wheel flat defects at different operating conditions using vibration signal processing. Signals were recorded by a data logger with a sampling rate of 10 kHz,

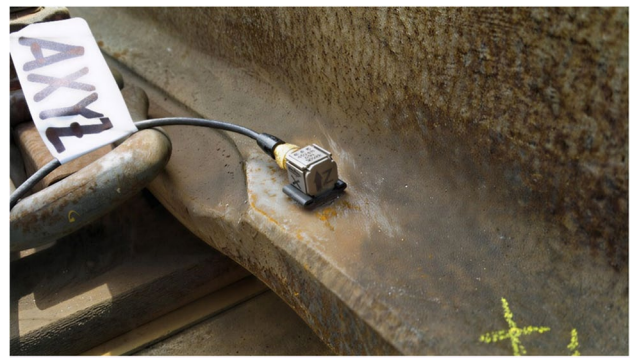


Fig. 13 Three axial piezoelectric accelerometer [155]

with a setup consisting of five piezoelectric accelerometers mounted vertically on the rail foot at the midspan between two adjacent sleepers. Avsievich et al. [158] investigated the stress–strain behavior of a railway track using high-precision accelerometers. The measurement system consisted of six three-axis digital accelerometers (ACt90) mounted directly on the rail as illustrated in Fig. 14. The setup enabled recording rail acceleration in three directions during the train passages, providing the determination of rail displacement and corresponding stresses through numerical modeling.

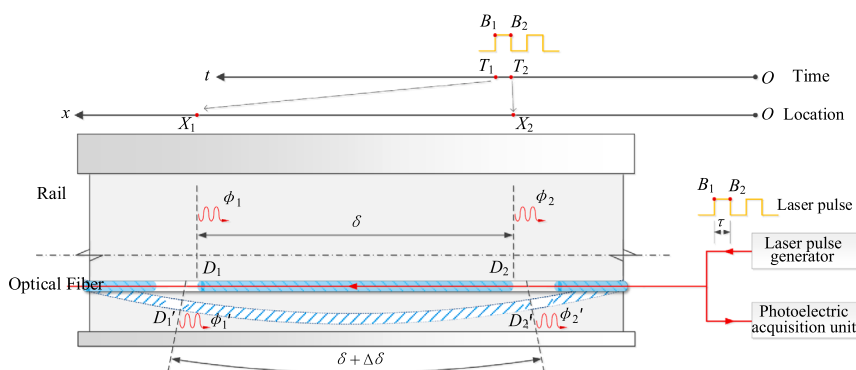


Fig. 12 Rail longitudinal strain measurement based on DAS: sensing principle and experimental measurement [152]

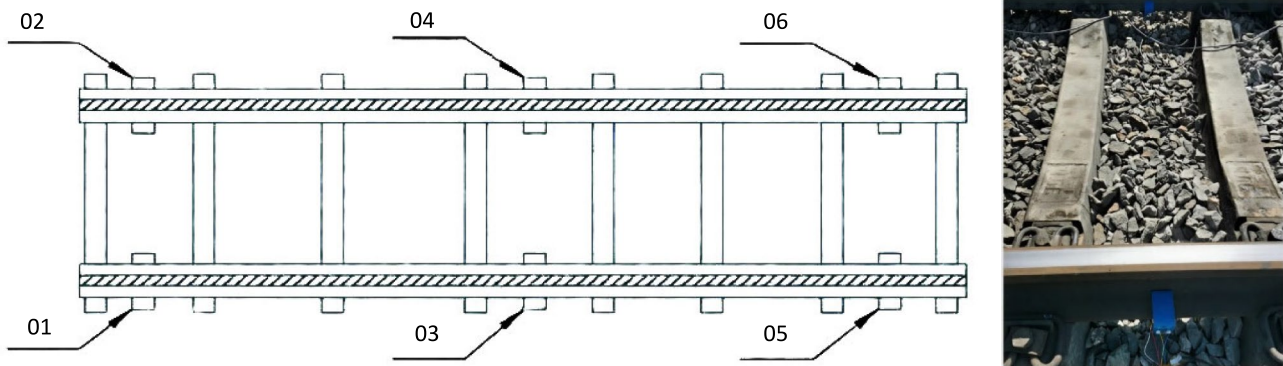


Fig. 14 Acceleration measurement layouts adopted in field tests by Avsievich et al. [158]

5.2.3 Ultrasonic sensors

Ultrasound has been frequently used to detect faults in the wheel tread. An electromagnetic acoustic (EMAT) or piezoelectric transducer is triggered when the wheel passes over it, generating a Rayleigh wave that travels around wheel's rolling surface [159]. Ultrasonic sensors provide real-time data on rail and wheel conditions, enabling immediate response to detected defects. Advanced algorithms process the ultrasonic signals to identify and quantify defects. This data is then transmitted to central monitoring systems for analysis and action.

Brizuela et al. [160, 161] investigated two alternative ultrasound techniques for detecting and quantifying wheel flats. In the first approach [161], they detected velocity changes at the contact point caused by the Doppler effect, using a monochromatic wave. Flats were detected by analyzing the variations of the Doppler frequency shift. In a second approach [160], ultrasonic pulses (Rayleigh waves) are sent over a measuring rail, as illustrated in Fig. 15a. In this case, two 1 MHz piezoelectric transducers were designed to emit and receive surface ultrasonic waves. The variations in the round-trip time of flight (RTOF) of the ultrasound pulse to the rail–wheel contact point allow for the detection and quantification of wheel flats.

Severe noise can obscure useful signals, making it difficult to identify defects in the echo signal. To improve the signal-to-noise ratio, Sun et al. [162] used piezoelectric transducers to detect faults in moving train wheels (Fig. 15b) and proposed an improved threshold function to address the signal discontinuity caused by non-stationary characteristics of ultrasonic detection signals. Moreover, to ensure good contact between the probe and the wheel, Montinaro et al. [163] proposed a non-contact ultrasonic technique for train wheel inspection that uses a laser interferometer to receive the ultrasonic wave without contact.

5.2.4 Acoustic sensors

Acoustic techniques employ microphones or acoustic emission (AE) sensors to capture sound waves generated by train–track interaction, enabling detection of anomalies and defects in various components. These techniques are primarily used for the wayside evaluation of wheels and axle bearings, particularly for identifying faults at the early stages. Trackside acoustic detectors (TADs) use arrays of microphones to record the noise produced by the bearings (see Fig. 16a). The operational frequency range of microphones in such arrays typically spans 20–40 kHz [164]. A key limitation of this technique is its susceptibility to environmental noise, as it relies on airborne sound transmission. Early stage faults often produce low-amplitude signals, resulting in a low signal-to-noise ratio that complicates reliable detection. Various approaches have been employed to address this issue, including directional microphones [165–167] or dedicated microphone arrays [168, 169].

The advantage of AE sensors (Fig. 16b) is their capability to detect very low-energy signals caused by bearing failures at an early stage or during slow-speed operation [170]. The frequency response of AE sensors typically ranges from 20 kHz to 1 MHz, offering a higher signal-to-noise ratio compared to more common vibration measurements. However, with these higher sampling rates, it is difficult to analyze the AE signal due to the huge amount of data in the collected time series. Amini et al. [37] used a wayside high-frequency acoustic emission system based on a set of AE sensors mounted on the rails, connected to a data acquisition unit and a data processor module with a sampling rate of 500 kHz. Bollas et al. [171] performed acoustic emission trials for online inspection of train wheels using several combinations of AE sensors (with a frequency range of 30–500 kHz) mounted on the rail web using magnetic hold downs (as shown in Fig. 16b).

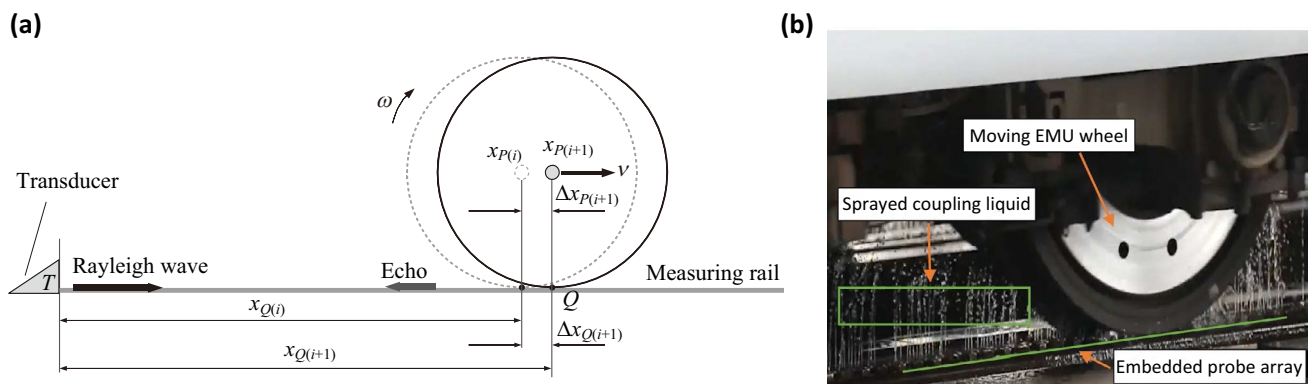


Fig. 15 Applications using ultrasonic techniques: **a** measuring arrangement [160]; **b** field test of ultrasonic diagnosis [162]

5.2.5 Laser sensors

Laser is the most widely used technique for accurately and regularly measuring wheel dimensions, including the wheel diameter and geometric tread parameters. It provides non-contact measurement with high-speed detection capability, making it ideal for analyzing high-speed movements and vibrations. Two main methods have been used for field measurements: laser displacement transducers (LDT) and structured light vision transducers (SLVTs).

An LDT is a device that uses a laser to measure the wheel diameter dynamically. The LDT measures the distance between the laser sensor and the wheel tread, and this distance is used to calculate the wheel diameter using a simple procedure [172]. Zheng et al. [173] developed a simplified method for measuring the diameter of train wheels precisely (with an error less than 0.3 mm) and dynamically by utilizing three one-dimensional laser displacement transducers (1D-LDTs), as shown in Fig. 17a.

These sensors are mounted on the rail and called S1, S2, and S3, respectively.

More recently, with the development of image-processing technology, SLVTs have become a commonly used method for on-site measurement of wheel tread profiles. These sensors use laser light to project a known pattern of pixels (a singular line, horizontal lines, crosshair, grid) onto the wheel surface. The way that these deform when striking surfaces allows vision systems to calculate the depth and surface information of the objects. To measure the geometrical parameters of wheel tread dynamically, Ran et al. [174] developed a measurement system based on a line-structured light vision sensor, as shown in Fig. 17b. In the system, four line-structured light sensors are mounted on the two sides of each rail to measure the wheel tread profile. Compared with a single-line laser, a multi-line laser can obtain 3D information about the measured object without repeated operations, which improves efficiency and accuracy. Sun et al. [175] proposed a framework for reconstructing symmetrical 3D

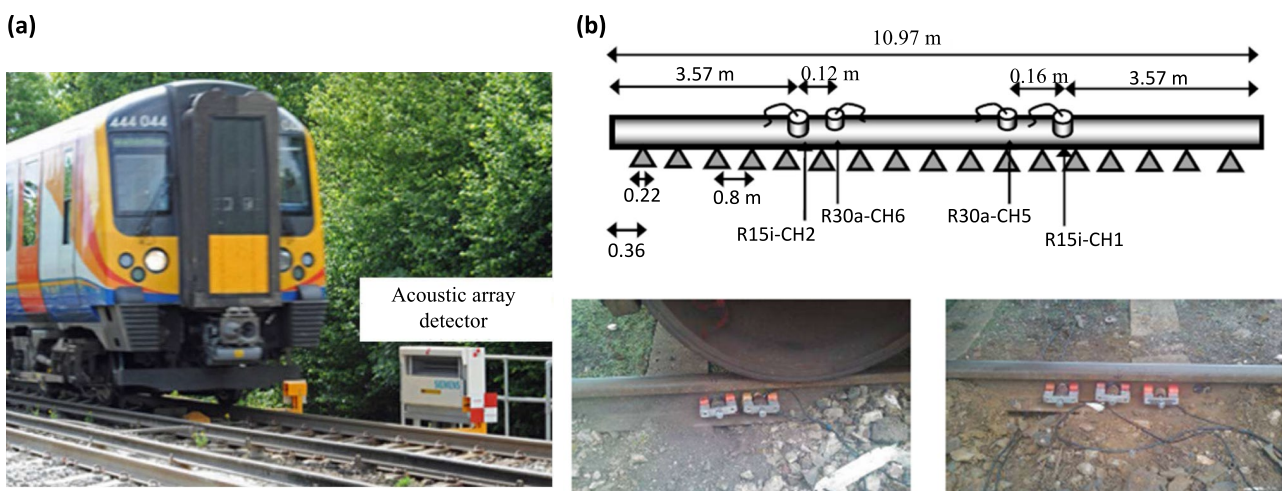


Fig. 16 Wayside acoustic measurements using: **a** microphone arrays [164]; **b** acoustic emission sensors [171]

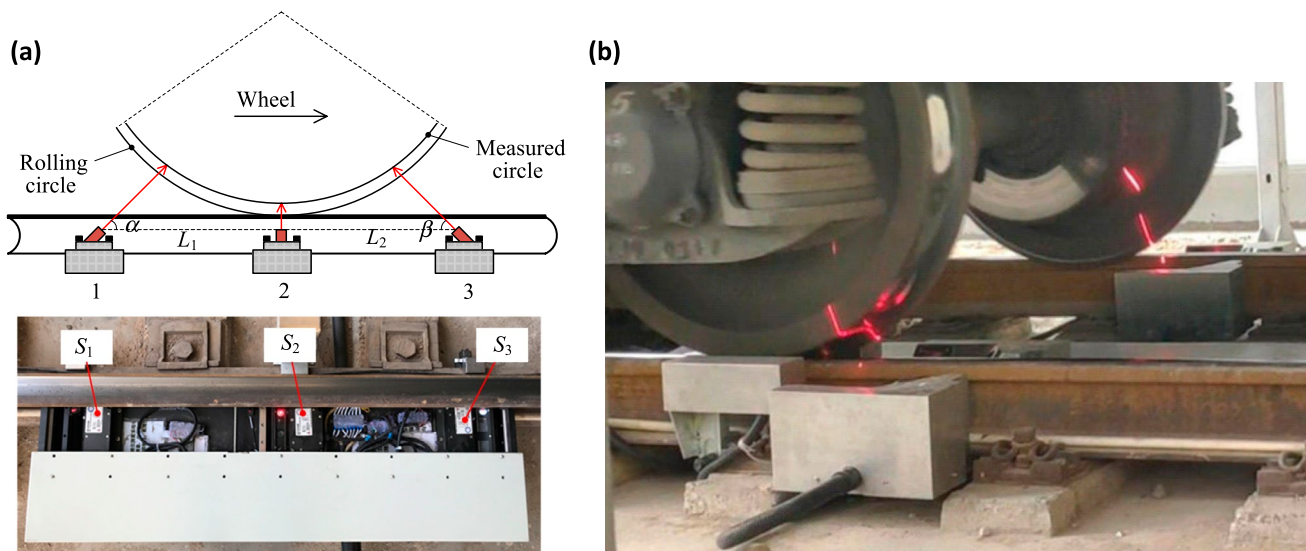


Fig. 17 Wheel profile measurements with laser techniques: **a** using three 1D-LDTs [173]; **b** using four line-structured light sensors [174]

revolving structures using general 3D profiles of the section obtained by a multi-line-structured light-vision sensor.

5.2.6 Thermal and RGB cameras

Wayside thermal and visible-spectrum cameras are also an automatic wheel and axle bearing inspection option. Some defects generate a characteristic heat pattern in the wheel or bearing that is clearly visible in thermal images. Other defects, however, may not exhibit a distinct thermal pattern but can be identified from visible-spectrum images, where texture and color are critical features. Recently, many researchers have investigated the possibility of detecting defects using images based on machine learning through computer vision methods that enable computers to extract information from images.

Regarding thermal images, Merainani et al. [176] developed an approach to detect hot boxes in infrared images captured by two types of wayside cameras. Kim et al. [177] used an infrared camera to monitor the underbody components of railway cars (traction motors and reducers) and used cross-correlation to detect abnormal heat. In the study of Deilamsalehy et al. [21], a computer vision method is developed for automatically detecting the sliding wheels from images taken by wayside thermal cameras. As a byproduct, the process also presents a method for detecting hot bearings from the same images. Jakubek et al. [178] proposed a thermovision-based methodology for assessing the thermal behavior of rolling bearings using a high-resolution infrared camera, enabling non-contact temperature measurement of individual bearing components (Fig. 18).

Many authors propose using high-resolution visible-spectrum cameras to capture images or videos of the wheel treads as trains pass by. Advanced algorithms and software then analyze the captured images to detect and measure defects and wear patterns on the wheel treads. Emoto et al. [179] proposed an automated system for inspecting and monitoring the wheels' condition by utilizing laser sensors and cameras for image data. It combined a laser sensor and image data captured using a digital video camera with a 640×480 -pixel resolution. In another study, Li et al. [180] proposed a virtual point tracking method to detect relative wheel-rail lateral displacement, which is an important physical quantity to evaluate hunting stability. A charge-coupled device (CCD) camera was used, positioned underneath the bogie of the train to capture the contact area between the wheelset and the rollers, as depicted in Fig. 19. The sampling frequency of the video was 25 frames per second, and the resolution of the captured image was 352×288 pixels.

Table 2 presents a synthesis of the main types of sensors used in wayside monitoring systems, categorized by measurement technique, sensor type, detection targets, and their technical principles (i.e., contact or non-contact operation), to enhance clarity and practical relevance.

6 Knowledge extraction from wayside systems

This section focuses on how wayside systems are used to monitor train wheel conditions and vehicle loads, emphasizing extracting meaningful insights from the collected data. These systems use advanced sensor technologies to assess

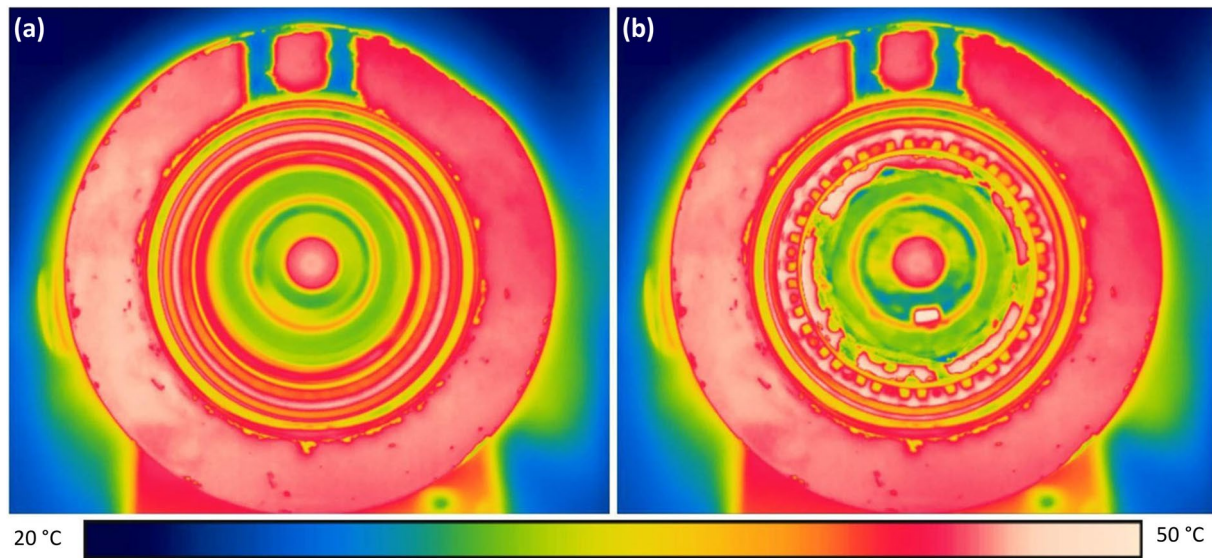


Fig. 18 An example of thermograms: **a** rotating bearing; **b** non-rotating bearings [178]

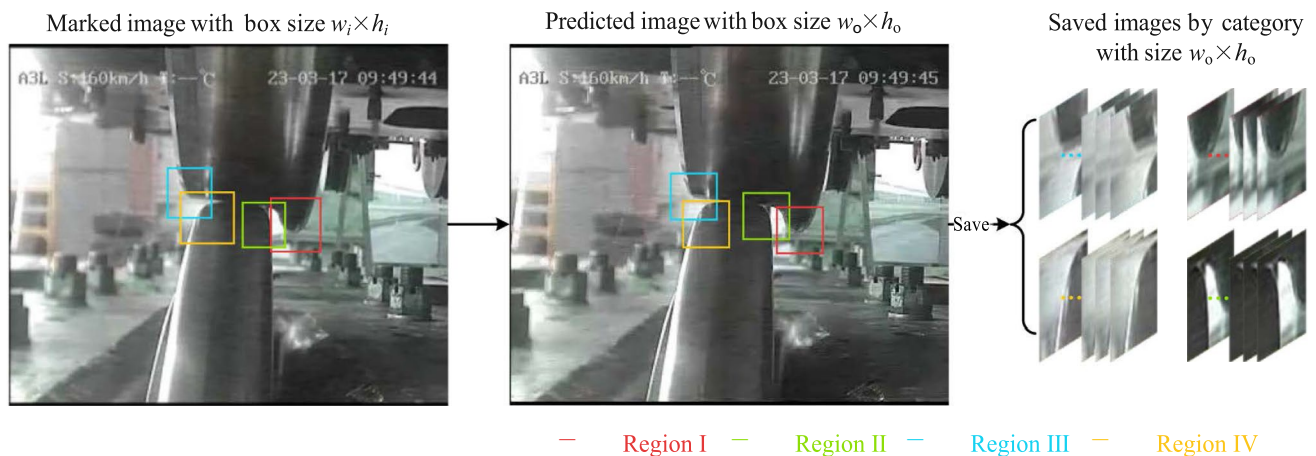


Fig. 19 Wheel–rail contact images provided by a CCD camera [180]

the condition of railway components in real time, enabling the identification of defects such as wheel flats, bearing faults, and unbalanced loads. By integrating machine learning and data analysis, wayside systems transform raw monitoring data into actionable knowledge, driving safety and operational efficiency improvements.

6.1 Wheel flats

Several advanced techniques have been developed to identify, localize, and assess the severity of wheel flats. These methods are categorized into signal processing-based approaches, machine learning-enhanced systems, and hybrid models. Belotti et al. [156] developed a methodology based on the discrete wavelet transform (DWT), which effectively

isolates defect signals from background noise using a single-sensor setup. Yang et al. [211] examined artificial defects in Chinese high-speed railways, detecting flats using test rods in close contact with wheel rims through Wavelet analysis. Nowakowski et al. [198] implemented envelope analysis using the Hilbert transform to detect wheel flats on trams, based on the processing of vibration signals in the time and frequency domains. Barman et al. [212] applied the wavelet transform (WT) and the Wigner–Ville transform to detect the presence of wheel flats. Mosleh et al. [188, 213] performed 3D numerical simulations based on a train–track interaction model using 12 strain gauges placed on the rail web to compute shear forces for identifying wheel flats. Gonçalves et al. [200, 214] utilized amplitude differences in the envelope spectrum for identifying wheel defects. Mosleh et al. [79, 80,

Table 2 Overview of the most common sensors used in wayside systems

Measurement technique		Sensors	Technical principles (contact/non-contact)	Detection targets	References
Deformation	Strain gauges		Contact	Wheel tread defects, train's overload, unbalanced loads, and hunting instability	Stratman et al. (2007) [145]; Milkovic et al. (2013) [181]; Lunys et al. (2016) [182]; Krummenacher et al. (2018) [146]; Alemi et al. (2019) [183]; Cortis et al. (2020) [184]; Zhou et al. (2020) [185]; Zhou et al. (2020) [186]; Ni et al. (2021) [187]; Mosleh et al. (2021) [188]; Pintao et al. (2022) [189]; Peng et al. (2022) [190]; Lourenço et al. (2023) [191]; Silva et al. (2023) [81]
					Filigrano et al. (2012) [149]; Wei et al. (2012) [192]; Liu et al. (2019) [148]; Ni et al. (2021) [187]; Mishra et al. (2022) [150]; Lan et al. (2023) [193]; Suhaime et al. (2023) [194]
					Peng et al. (2014) [153]; Gao et al. (2022) [152]; Zhang et al. (2022) [154]
Vibration	Laser displacement transducers (LDT)	Distributed acoustic sensing (DAS)	Contact	Wheel tread defects, axle bearing faults, and hunting instability	Zheng et al. (2019) [173]; Gao et al. (2020) [195]; Liu et al. (2021) [196]
					Skarlatos et al. (2004) [197]; Belotti et al. (2006) [156]; Nowakowski et al. (2019) [198]; Mosleh et al. (2023) [199]; Mohammadi et al. (2023) [27]; Guedes et al. (2023) [76]; Gonçalves et al. (2023) [200]; Cai et al. (2023) [201]; Salehi et al. (2023) [157]; Jorge et al. (2024) [30]
					Salzburger et al. (2009) [159]; Montinaro et al. (2019) [163];
Ultrasonic		Electro-magnetic acoustic (EMAT)	Non-contact	Wheel tread defects	Brizuela et al. (2010) [161]; Brizuela et al. (2011) [160]; Sun et al. (2021) [162]
Acoustic		Piezoelectric transducers	Contact	Wheel tread defects	Bollas et al. (2013) [171]; Amini et al. (2016) [37]; Aktas et al. (2018) [202]; Chang et al. (2022) [203]
		Acoustic emission (AE) sensors	Contact	Axle bearing faults and wheel tread defects	Dybala et al. (2013) [204]; Hu et al. (2013) [205]; Liu et al. (2017) [167]; Chen et al. (2020) [168]; Komorski et al. (2021) [169]; Kukenas et al. (2021) [165]
		Microphones	Non-contact	Wheel profile defects	Gao et al. (2014) [172]; Zheng et al. (2019) [173]; Liu et al. (2021) [196];
Laser		1D laser displacement transducers (1D-LDT),	Non-contact	Wheel profile defects	Cheng et al. (2016) [206]; Xing et al. (2016) [207]; Pan et al. (2018) [208]; Sun et al. (2021) [175]; Ran et al. (2021) [174]
		Structured light vision transducers (SLVT)	Non-contact	Wheel profile defects	

Table 2 (continued)

Measurement technique	Sensors	Technical principles (contact/non-contact)	Detection targets	References
Image	Thermal camera	Non-contact	Axle bearing faults seized/sliding wheels	Kim et al. (2014) [177]; Deilamsalehy et al. (2017) [21]; Tarawneh et al. (2020) [209]; Merainani et al. (2023) [176]
	High-speed camera	Non-contact	Wheel profile defects	Pan et al. (2019) [210]; Emoto et al. (2024) [179]; Li et al. (2024) [180]

[199] and Mohammadi et al. [27] integrated AI techniques with a wayside monitoring system to detect flats using outlier analysis, enhancing defect detection accuracy. In another study, Mosleh et al. [78, 84] not only detected wheel flats but also classified their severity using the k -means clustering approach. Mohammadi et al. [31] advanced wheel flat identification by enabling damaged wheel localization through a hidden Markov model (HMM), while a sparse autoencoder and k -means clustering were implemented for defect severity classification, considering both single and multi-defective wheels. Cao et al. [215] proposed a two-step approach for analyzing rail force time-history to measure flats based on significant impact forces. Such ML-driven methods enhance detection accuracy, streamline maintenance, and lower costs. Jorge et al. [30] employed a sparse autoencoder and k -means clustering technique to detect, diagnose, and classify wheel flats. Emerging technologies and hybrid approaches are bridging gaps in traditional methods.

Alemi et al. [216] used a data fusion approach to reconstruct wheel defect signals from wheel-rail contact signals for damage detection. Gao et al. [217] introduced a parallelogram system for flat detection using a waveform-based analysis technique, though high speeds and vibrations posed challenges to accuracy. Gao et al. [195] further developed a laser-based detection system that links flat dimensions to the amplitude of the impact signals, while Zhou et al. [185] utilized refined rail response models for precise defect localization based on the variation in strain signal strength. Optical systems by Filograno et al. [149] employed trackside sensing to detect periodic signals from flat wheels, counting axles and monitoring dynamic loads in high-speed trains through the repetition frequency of defect impacts related to the train speed and wheel diameter. Similarly, Mishra et al. [150] developed an algorithm to generate a reflective index to monitor wheel conditions by measuring the shift in the wavelength of the reflected light.

Aktas et al. [202] explored an in-operation method for detecting wheel flats that employs parametric constraint optimization, ensuring its functionality is independent of vehicle speed. This method relies on a noisy, periodic, impulsive signal influenced by train speed and wheel diameter. Komorski et al. [169] used Fourier and Hilbert transforms combined with spectrum analysis to detect tram wheel flats based on acoustic measurements. Chang et al. [203] proposed a detection method using an improved synthesize health index (ISHI) integrated with a time-adaptive threshold (time-ATH) to detect damaged signals and effectively minimize interference caused by wheel-rail rolling. Validation using real-world datasets demonstrated enhanced accuracy and detection rates compared to existing methods.

Thermal imaging systems by Deilamsalehy et al. [218, 219] combined wavelet decomposition and Hough transform for identifying wheel flats and bearing defects under

complex conditions. These hybrid methods integrate the strengths of signal processing and ML, though external factors such as speed and environmental noise remain challenging. Additional studies contribute unique perspectives to wheel defect detection. Gordan et al. [23] developed a blockchain-based, data-driven approach for pre-condition monitoring aimed at improving the reliability of measured dynamic responses caused by wheel flats in freight trains. For this purpose, a data mining process is designed to create a local blockchain.

A comparative analysis of these approaches highlights key trends. Signal processing techniques provide cost-effective solutions suitable for routine monitoring, while ML and hybrid methods offer enhanced accuracy and predictive capabilities, making them ideal for modern railways. A summary of the main references discussed in this section regarding the identification, localization, and diagnosis of wheel flats in wayside condition monitoring is presented in Table 3. The table highlights important aspects such as dataset type (E—experimental,

N—numerical, or A—artificial), train type, and the main techniques employed, thereby consolidating the presented information more effectively.

6.2 Polygonization

Polygonal wheels pose significant challenges for railway operations, contributing to excessive vibrations, noise, accelerated wear on infrastructure, and increased maintenance costs. Early detection and accurate identification of these defects are essential to mitigate their impact and ensure stable and efficient train operations. Several researchers have addressed the challenge of detecting polygonal wheels through various techniques, focusing on both time- and frequency-domain analyses.

Alemi et al. [183] proposed a monitoring system to identify high-impact forces and various wheel conditions, including wheel out-of-round (OOR) defects. Their study employed a data fusion process to reconstruct informative signals from discrete samples, leading to more

Table 3 Summary of studies on wheel flats identification using wayside systems

References	Data set type	Train type	Techniques/methods
Belotti et al. 2006 [156]	E	Passenger	Detection and localization: power spectral density
Yang et al. 2016 [211]	E/A	Passenger	Detection: wavelet analysis-based method
Nowakowski et al. 2019 [198]	E	Tram	Detection: envelope spectrum analysis
Barman et al. 2020 [212]	E	Passenger	Detection: WT and Wigner–Ville transform
Mosleh et al. 2021 [188, 213]	N	Passenger/freight	Detection: envelope spectrum analysis
Gonçalves et al. 2023 [200]			
Mosleh et al. 2021, 2023, Mohammadi et al. 2023 [27, 77, 79, 80, 199]	N	Passenger/freight	Unsupervised detection: outlier analysis
Mosleh et al. 2022, 2023 [78, 84]	N	Passenger	Unsupervised detection: outlier analysis Unsupervised classification: k -means
Ghiasi et al. 2024 [220]	N	Passenger	Unsupervised multistage clustering including k -means and DBSCAN
Mohammadi et al. 2025 [31]	N	Freight	Unsupervised detection: outlier analysis Unsupervised localization: HMM Unsupervised classification: autoencoder and k -means
Salehi et al. 2023 [157]	E	Motor car	Detection: multi-channel singular spectrum analysis (MSSA)
Lourenço et al. 2023 [191]	N	Freight	Unsupervised diagnosis: isolation forest and HMM
Jorge et al. 2024 [30]	N	Passenger	Detection: outlier analysis diagnosis and classification: k -means
Gao et al. 2019 [217]	E	–	Detection: parallelogram mechanism (waveform analysis)
Zhou et al. 2020 [185]	N/E	Passenger	Detection and localization: data fusion
Filigrano et al. 2012 [149]	E	Passenger	Detection: wavelength variation analysis
Mishra et al. 2022 [150]	N	Passenger	Detection: FFT
Aktas et al. 2018 [202]	E	Passenger	Detection: peak detection technique
Komorski et al. 2021 [169]	E	Tram	Detection: envelope spectrum analysis
Chang et al. 2022 [203]	E	–	Detection: data fusion
Deilamsalehy et al. 2016 [218]	E	–	Supervised detection: SVM
Deilamsalehy et al. 2017 [219]	E	–	Detection: Hough transform
Berizuela et al. 2011 [160]	E	–	Detection: round-trip time of flight (RTOF)
Gordan et al. 2024 [23]	N	Freight	Detection: blockchain, data mining

precise detection and classification of wheel defects. This approach demonstrates the potential for minimizing maintenance disruptions by providing actionable insights into wheel conditions. Guedes et al. [76] developed a machine learning methodology for automatically detecting polygonal wheels on freight trains using synthetic vibration data and outlier analysis. Moreover, a sensitivity analysis was performed to evaluate the influence of sensor quantity and placement on the detection accuracy. Acceleration measurements were collected at a sampling frequency of 10 kHz from 8 positions—6 located on the rail and 2 on the sleepers. The inclusion of this sensitivity analysis underscores the importance of optimizing sensor configuration for effective defect detection and provides valuable recommendations for practical implementations.

Based on their earlier work, Guedes et al. [8] developed an algorithm capable of classifying polygonal wheels based on harmonic order and defect amplitude, using a *k*-means clustering technique. This enhanced methodology provided granular insights into the severity of polygonization, offering a robust tool for targeted maintenance strategies. Lourenço et al. [191] proposed a wayside monitoring system based on machine learning, using vibration analysis to diagnose polygonal wheels. This system employed the Isolation Forest method for defect detection and classification, while a HMM accurately identified the position of the defective wheel. The combination of these methods allows for both precise defect detection and localization, which is critical for improving maintenance efficiency. Jorge et al. [30] advanced the field using sparse autoencoder and *k*-means clustering techniques for the detection, diagnosis, and classification of polygonal wheels. This approach combined dimensionality reduction and clustering, enabling improved accuracy and computational efficiency in defect classification. A comparative evaluation of this method against other existing techniques, particularly in terms of detection rates and computational speed, would further highlight its advantages.

Table 4 provides a comprehensive summary of the key references covered in this section concerning the identification and diagnosis of polygonal wheels in wayside condition monitoring. The presented information includes dataset type (N—numerical), train type, and the primary techniques/methods used.

6.3 Shelling and cracks

The detection of shelling and cracks in train wheels is crucial for ensuring operational safety and minimizing the risk of derailments or other failures caused by wheel defects. Several studies have focused on this issue, proposing various methods and systems to assess the health of train wheels. Rombach et al. [221] proposed an unsupervised method for railway wheel defect detection. Their approach implemented contrastive learning to group similar nearby samples and separate distant ones, followed by a one-class SVM (OC-SVM) for detecting anomalies. In a subsequent study, Rombach et al. [222] implemented OC-SVM for enhanced crack and shelling detection. The use of unsupervised learning techniques highlights the adaptability of machine learning methods in addressing the complexities of wheel defect detection. Peng et al. [223] introduced a filtering algorithm to improve the signal-to-noise ratio (SNR) and defect detection rate. The algorithm increased the amplitude of the defect echo and decreased the amplitude of the interference echo, thereby enhancing the system's ability to accurately detect defects. Zhang et al. [224] proposed a deep learning algorithm to improve upon existing YOLO (you only look once) models by incorporating advanced features like Swin-Transformer and BiFPN for detecting small cracks. In addition, a modified loss function called LEIoU was developed to enhance the accuracy of bounding box regression. Table 5 summarizes the main references discussed in this section related to shelling and cracking detection using wayside condition monitoring. It details the type of dataset (E—experimental or N—numerical), railway vehicle types, and the primary methodologies employed to facilitate a clearer understanding of the research.

Table 4 Summary of studies on polygonal wheel identification using wayside systems

References	Data set type	Train type	Techniques/methods
Alemi et al. 2019 [183]	N	—	Detection and identification: data fusion: single and multi-sampling method
Guedes et al. 2023 [76]	N	Freight	Unsupervised detection: outlier analysis
Guedes et al. 2024 [8]	N	Freight	Unsupervised detection: outlier analysis Unsupervised classification: <i>k</i> -means
Lourenço et al. 2023 [191]	N	Freight	Unsupervised diagnosis: isolation forest, HMM
Jorge et al. 2024 [30]	N	Passenger	Unsupervised detection: outlier analysis Unsupervised diagnosis and classification: SAE, <i>k</i> -means

Table 5 Summary of studies on shelling and crack wheel detection using wayside systems

References	Data set type	Train type	Techniques/methods
Stratman et al. 2007 [145]	E	Freight	Detection-prediction: peak impact reading
Rombach et al. 2025 [222]	E	Passenger	Unsupervised detection: OC-SVM
Peng et al. 2016 [223]	E/N	Passenger	Detection: filtering-based algorithm
Zhang et al. 2024 [224]	E	–	Supervised deep learning detection: UT-YOLO algorithm

UT stands for ultrasonic testing

6.4 Bearing defects

Several researchers have proposed diverse methodologies employing acoustic data, thermal imaging, and advanced signal processing techniques for bearing defect detection. These methods aim to identify early signs of bearing faults, enable condition-based maintenance, and reduce the risk of failures. Cline et al. [225] identified specific frequencies within the acoustic signature that correspond to different types of bearing defects on cargo trains based on the bearing geometry and the train speed. Building on this, Zhang et al. [226] and Hu et al. [205] demonstrated an innovative approach that uses the Doppler Shift to enhance signal processing reliability for bearing fault diagnosis. Their study involved signal extraction, acoustic source analysis, and the design of noise-filter algorithms using the crazy climber detection algorithm and Doppler-let transform and resampling. These methods have significantly advanced the field by mitigating Doppler effects in dynamic railway environments.

Papaelias et al. [164] implemented techniques such as moving root mean square (RMS) and windowed FFT to analyze defective signals. Their results indicated that the presence of resonance frequencies could effectively indicate a bearing defect. Liu et al. [167, 227] introduced iterative algorithms and ensemble empirical mode decomposition (EEMD) to further reduce Doppler effects and enhance diagnostic accuracy. In a complementary study, Zhang et al. [228] utilized time–frequency data fusion and envelope spectrum analysis to suppress Doppler distortion and background noise, yielding improved diagnostic clarity.

Moreover, Amini et al. [37] applied time spectral kurtosis (TSK) analysis on high-frequency acoustic signals for railway axle-bearing fault detection. Their findings emphasized the superiority of TSK over conventional methods. Similarly, Wang et al. [229] developed a double-searching solution based on an improved Doppler-let transform for detecting damaged bearings. Zhang et al. [230] designed a time-varying spatial filtering rearrangement scheme to enhance multi-bearing fault diagnosis. In another study, Zhang et al. [231] employed a combination of improved singular value decomposition and resonance-based signal sparse decomposition for effective fault feature extraction,

while Zhang et al. [232] used spline-kernel methods to improve damaged bearing detection accuracy after Doppler reduction. Spline kernels, which employ smooth, piecewise-polynomial functions, enable precise modeling of complex relationships within the measured data. Liu et al. [233] suggested a two-stage learning model for diagnosing bearing faults, addressing the effect of Doppler distortion on signals. Liu et al. [234] proposed a kurtosis and energy estimator-based approach to enhance weak axle-bearing signals. Hu et al. [235] applied wavelet domain moving beamforming (WMB) to improve fault diagnosis in axle bearings, while Shi et al. [236] introduced an advanced feature extraction method using linear discriminant analysis to detect bearing damage effectively.

Thermal imaging has also been explored for detecting bearing faults. Deilamsalehy et al. [218] used thermal cameras and visual images to extract features using a histogram of oriented gradients and classified faults with a support vector machine. Their subsequent study [219] combined wavelet decomposition and Hough transform for feature extraction, fusing thermal and visual images for enhanced defect detection.

Infrastructure-level solutions have also been proposed. Vale et al. [124] developed the OPTIBOX software tool, to assist infrastructure managers in optimizing the placement of hot axle box detectors (HABD) within railway networks. This tool not only identifies the optimal locations for detectors but also minimizes their required number. Merainani et al. [176] introduced a system combining artificial intelligence with the Kalman filter to detect, count, and track moving train wheel axles using thermal infrared images. Dias et al. [237] combined multi-layer perceptron (MLP) and an adaptive filter (set membership) to detect and predict hot bearings (hot box detector). Vale et al. [238] further enhanced maintenance strategies with a software tool that optimizes schedules by detecting axle-bearing failures at an early stage, enabling condition-based maintenance. Yang et al. [239] designed a multi-objective optimized wavelet filter for extracting repetitive transients from bearing signals, considering both impulsiveness and cyclo-stationarity. These techniques collectively represent state-of-the-art advancements in bearing defect detection and monitoring.

Table 6 Summary of studies on wheelset bearing detection using wayside systems

References	Data set type	Train type	Techniques/methods
Cline et al. 1998 [225]	E	Freight	Supervised detection: neural network
Papaelias et al. 2014 [164]	E	Freight	Defect identification: moving RMS
Liu et al. 2016 [227]	E/N	–	Detection: EEMD
Liu et al. 2017 [167]	E/N/A	–	–
Zhang et al. 2016 [228]	E/N	–	Detection: envelope spectrum analysis
Amini et al. 2016 [37]	E/N	Freight	Detection: TSK
Liu et al. 2021 [233]	E/N	Freight/ passenger	Supervise detection: SVM
Liu et al. 2023 [234]	E/N	Freight/ passenger	Supervised detection: SSD-KE, envelope spectrum analysis
Hu et al. 2022 [235]	E/N	–	Detection: WMB
Shi et al. 2022 [236]	E/N	–	Supervised detection: extreme learning machine (ELM)
Deilamsalehy et al. 2016 [218]	E	–	Supervised detection: SVM
Deilamsalehy et al. 2017 [219]	E	–	Detection: Hough transform
Vale et al. 2020 [124]	N	–	Detection and optimization: OPTIBOX tool
Merainani et al. 2023 [176]	E	Freight/ passenger	Supervised detection: YOLO-v4
Dias et al. 2024 [237]	E	–	Supervised detection and prediction: MLP
Vale et al. 2014 [238]	E	Freight/ passenger	Unsupervised detection: threshold-based
Yang et al. 2020 [239]	A/E	High-speed freight locomotive	Detection: kernel density estimation (KED)

Table 6 summarizes the key references explored in this section concerning bearing defect identification and prediction using wayside condition monitoring. This table highlights critical elements, including the type of data used (E–experimental, N–numerical, or A–artificial/simulated), the specific types of railway vehicles assessed, and the main techniques implemented.

6.5 Dynamic loads, overloading, and unbalanced loads

Proper monitoring of railway vehicle load is critical to ensuring safe service and operation. Pan et al. [240] employed photoelectric conversion and wavelength demodulation to obtain strain signals. The proposed technique calibrated the sensitivity coefficient between FBG wavelength offset and axle load and combined it with real-time offset data for dynamic load detection. Afterwards, axle load distribution across tracks on both sides was compared to detect unbalanced carriages. Similarly, Lan et al. [193] used optical sensing to analyze the relationship between rail strain and track slab strain in both time and frequency domains. Their findings revealed that vertical strain levels in the track slab are lower under high-speed train dynamic loads, offering insights into load distribution and structural behavior. Allotta et al. [241] introduced an algorithm employing weighted least square optimization (WLSO) and non-linear least square optimization (NLSO) to estimate train parameters like axle positions and loads. Their method uses rail and sleeper-ballast deformation data to deliver precise linear approximations and superposition analyses.

Mosleh et al. [242] examined various factors influencing static load estimation, including train speed, rail pad stiffness and damping, foundation stiffness and damping, and track unevenness. These findings provided foundational insights for dynamic load estimation methodologies. Building on earlier studies, Mosleh et al. [243, 244] proposed a weigh-in-motion algorithm to estimate static load using strain gauge measurements while the train is in motion. This innovative approach was validated by Pintão et al. [189, 245] using synthetic data from dynamic wayside weighing systems, consisting of six pairs of strain gauges installed along the rail. Their findings confirmed the robustness of WIM algorithms for real-world applications.

Silva et al. [81] developed an unsupervised ML-based methodology for automatically detecting and classifying longitudinal and transversal unbalanced loads. Using outlier analysis and *k*-means clustering techniques, their research adhered to reference guidelines and employed synthetic data from eight strain gauges installed along the rail. This study underscored the potential of unsupervised learning in detecting anomalies in load distribution. Moreover, Vale and Calçada [246] further highlighted the impact of track irregularities on vehicle–track interaction responses, which directly influence dynamic contact forces. Their research provided crucial insights into the interplay between infrastructure irregularities and load dynamics. Ding et al. [247] introduced two weigh-in-motion systems designed to monitor the loads of passenger and freight trains. These systems calculated critical parameters, including train geometry, axle number, axle load, train weight, and speed, using

Table 7 Summary of studies on dynamic loads, overloading, and unbalanced loads using wayside systems

References	Data set type	Train type	Techniques/methods
Lan et al. 2023 [193]	E	Passenger	Dynamic load monitoring: time-domain analysis
Allotta et al. 2014 [241]	E/N	Freight	Load estimation: WLSO and NLSO
Mosleh et al. 2017 [242]	N/A	–	Static load assessment: dynamic load distribution analysis
Mosleh et al. 2020 [243]	N/E	–	Load assessment: shear effort assessment
Mosleh et al. 2019 [244]	N	Freight	Static load assessment: threshold-based method
Ding et al. 2018 [247]	E	Passenger/freight	Detection: calibration method
Pintao et al. 2022 [189, 245]	A	Passenger/freight	Static load estimation: shear force measurement-based method
Silva et al. 2023 [81]	N	Freight	Supervised detection: outlier analysis Supervised classification: <i>k</i> -means

Table 8 Summary of studies on the identification of hunting movement using the wayside system

Citations	Data set type	Train type	Techniques/methods
Peng et al. 2022 [190]	E/N	Freight	Lateral motion monitoring: polynomial fitting
Wang et al. 2024 [248]	E/N	Freight	Lateral motion monitoring: diagnostic index

shear measurements obtained from trackside sensors. Their contribution represents a comprehensive approach to load monitoring, accommodating a diverse range of train types and operational conditions. The main references related to load estimation and unbalanced load identification using wayside condition monitoring are summarized in Table 7. This table outlines important aspects such as input dataset type (E–experimental, N–numerical, or A–artificial), type of trains, and techniques used, providing a clear overview of the research.

6.6 Hunting movement

To monitor vehicle hunting instability, Peng et al. [190] developed a method to detect lateral motion in railway systems using technologies like machine vision and eddy current sensors, which work based on the principle of electromagnetic induction. This system provides a reliable means of detecting early signs of instability, facilitating timely interventions. Wang et al. [248] proposed a hunting detection system specifically designed to evaluate hunting performance by measuring wheelset lateral displacement. Their method establishes a mapping relationship between the hunting coefficient and vehicle hunting state, offering a systematic way to assess and quantify instability. Table 8 summarizes key references on lateral motion monitoring with wayside condition monitoring, including data set types

(E–experimental or N–numerical), railway vehicle types, and the techniques developed.

6.7 Multiple damage scenarios

This section reviews studies focusing on the techniques for damage identification and wheel health assessment in rail vehicles, considering multiple damage scenarios. Asplund et al. [249] developed a wayside monitoring system capable of detecting out-of-roundness, including flats and spalling. This system also predicts the impact of speed on the force response of the defective wheel using the linear regression model (LRM). Building on similar efforts, Asplund et al. [34, 35, 250–252] proposed a wayside system using high-speed cameras and lasers based on optical sensing techniques to measure train wheel profiles and detect OOR defects. Krummenacher et al. [146] implemented machine learning techniques, such as SVM and convolutional neural network (CNN), to automatically identify various types of wheel defects, including wheels flats, shelling, and non-roundness defects during normal operation. Similarly, Wei et al. [192] proposed a system to measure track strain responses for wheel conditions assessment, providing a condition index to detect OOR defects, flat wheels, spalling, and shelling.

Liu et al. [148] proposed a trackside wheel tread defect detection system using outlier analysis with experimental data for passenger vehicles, specifically targeting OOR damages. Qing Ni et al. [187] advanced this approach by employing sparse Bayesian learning (SBL) and probabilistic analysis to detect OOR defects on wheel surfaces. Ran et al. [174] explored optical sensing-based methods for measuring wheel tread geometric parameters, including flange height and flange thickness, using a line-structured light-vision to detect defective flanges.

In metro systems, Kılınc et al. [253] developed a trackside monitoring system to detect wheel defects such as flats, shelling, and cracked wheels using vibration analysis combined with SVM (linear kernel), and Fisher linear discriminant analysis (FLDA). Expanding on this work, Kılınc et al. [254] implemented both acoustic and vibration

analysis techniques, using linear configuration pattern Kurtograms (LCP-K), SVM-I, decision tree, and linear perceptron (PERLC) to identify defects such as flat spots, shelling, cracked wheels and wheelset eccentricity.

Chen et al. [168] developed a time-domain beamforming method with kurtosis to detect and localize wheel defects, emphasizing its effectiveness in identifying dents, squats, and flats via acoustic-based trackside monitoring systems. Guo et al. [255] developed an image-based inspection technique for monitoring wheel tread defects, such as flats and shelling, regardless of the tread type, using SVM for defect classification. Similarly, Xing et al. [256] developed a machine vision method based on an enhanced YOLOv3 framework, incorporating convolution layers and feature pyramids for defect detection through wheel images.

Trilla et al. [257] developed a wayside system based on an image-processing technique using CNN to detect and diagnose tread defects, including spalls and flat spots. Furthermore, Sun et al. [162] applied a threshold denoising method for ultrasonic signals to identify defects in the electric multiple-unit (EMU) trains' wheels, specifically targeting through holes, flat bottom holes, and circumferential cracks. Table 9 provides the input dataset type (E—experimental, N—numerical, or A—artificial), the type of the trains, and the developed techniques to identify trains with multiple damages.

7 Challenges, gaps and future suggestions

The integration of wayside condition monitoring systems into the railway industry has significantly contributed to reducing component-related failures and improving operational safety. However, achieving widespread adoption

and realizing their full potential of WCM technologies still requires overcoming several technical, operational, and economic challenges. Key gaps persist in areas such as data quality, system interoperability, scalability, and the generalization of predictive models across different railway networks. In addition, comprehensive region-specific economic viability studies are essential to justify large-scale investments and deployment. This section highlights the main challenges, research gaps, and opportunities for future development, structured under distinct thematic subsections.

7.1 Current challenges in wayside sensor deployment and performance

Strain gauges and fiber Bragg grating (FBG) sensors need to be installed under the rails and become fixed once deployed, which significantly limits their portability and flexibility for dynamic monitoring applications. In addition, installing these sensors presents challenges, such as the need for precise positioning and careful alignment and often potential disruptions to rail operations during installation [258]. This lack of flexibility complicates maintenance and adaptation to changing monitoring needs.

In contrast, vision sensors offer a distinct advantage over strain gauges and FBG sensors, as they do not require installation under the track. Vision-based monitoring systems can be easily relocated and repositioned according to traffic conditions, facilitating more flexible monitoring. However, these systems require careful and repeated calibration to maintain accuracy, particularly when environmental conditions such as lighting or weather change.

Table 9 Summary of studies on the identification of wheel multi-damage using the wayside system

References	Data set type	Train type	Techniques/methods
Asplund et al. 2024 [249]	E	locomotive	Detection prediction: LRM
Krummenacher et al. 2018 [146]	N	Freight	Supervised detection and diagnosis: SVM and CNN
Wei et al. 2012 [192]	E	Passenger/freight	Detection: condition index (CI) based
Asplund et al. 2014 [250]	E/N	—	Detection: WPMS
Asplund et al. 2014 [251]			
Asplund et al. 2016 [34]			
Asplund et al. 2018 [252]			
Xiao Zhou Liu et al. 2019 [148]	E	Passenger	Detection: outlier analysis
Qing Ni et al. 2021 [187]	E	Passenger	Unsupervised detection: SBL
Ran et al. 2021 [174]	E	—	Detection: WPMS
Kılınc et al. 2017 [253]	E/A	Metro	Supervised diagnosis: SVM-I, FLDA
Kılınc et al. 2022 [254]	E	Metro	Supervised detection: LCP-K Diagnosis: SVM-I, Decision tree, and PERLC
Chen et al. 2020 [168]	E	—	Detection and localization: Kurtosis beamforming
Guo et al. 2017 [255]	E	—	Supervised detection: SVM
Xing et al. 2022 [256]	E	Metro	Supervised detection: YOLOv3
Trill et al. 2021 [257]	E	—	Supervised detection and diagnosis: CNN
Sun et al. 2021 [162]	E	Passenger	Supervised detection: Bayesian blind source separation, Diagnosis: deep neural networks

Ultrasonic and acoustic emission (AE) sensors are also utilized in the wayside inspection, but they present significant limitations. Their response speed is quite slow, requiring trains to decelerate for precise data acquisition [162]. In addition, these sensors are highly sensitive to ambient noise levels. In high-noise environments, such as those typical of operational railway lines, signal quality can be severely degraded, reducing the effectiveness of fault detection. High-speed cameras, especially when integrated with laser-based systems, provide a promising alternative by offering rapid, high-precision monitoring of wheel profiles and tread defects. These systems not only deliver reliable and accurate measurements but also offer substantial portability, facilitating the flexible deployment of wayside monitoring solutions [175]. Nevertheless, ensuring consistent image quality under different operational and environmental conditions remains a technical challenge that must be addressed to optimize their effectiveness.

Future research should focus on enhancing the durability, portability, and self-calibration capabilities of these sensor technologies. Developing wireless, self-powered, and easily reconfigurable sensor units could significantly expand the adaptability and resilience of WCM systems to meet evolving operational demands.

7.2 Current challenges in processing techniques

Signal acquisition from strain gauges, FBG sensors, ultrasonic transducers, and acoustic emission sensors is often significantly impacted by environmental and operational noise, which can mask or distort the fault signatures of interest [188]. In many cases, repetitive fault patterns are obscured, complicating reliable fault detection. Consequently, there is a strong need for advanced signal processing techniques integrated with ML methods to extract meaningful information under noisy conditions. Preprocessing steps, such as adaptive filtering or time–frequency analysis, could enhance the signal-to-noise ratio and improve feature extraction reliability.

Vision-based monitoring systems also present a unique set of challenges. Although they offer notable advantages, such as portability and early detection capability, detecting defects in moving targets like wheels introduces considerable difficulties. Surface defects often co-occur, making visual classification complex. Moreover, vision systems must cope with varying lighting conditions, reflections, shadows, and motion blur, all of which can degrade image quality [21]. To ensure effective operation, vision systems must also satisfy stringent requirements regarding the field of view, frame rate, and working distance to capture the complete circumference of fast-moving wheels.

Furthermore, real-time WCM applications face critical demands related to memory consumption, processing latency,

and data throughput. The massive amount of high-frequency data generated, particularly in high-speed rail environments, can overwhelm conventional data processing pipelines. Efficient algorithms are needed that can balance computational complexity with detection accuracy without compromising system responsiveness. Although ML and DL techniques show strong potential for automated defect detection, many existing models still struggle with robustness and reliability when faced with noisy, incomplete, or highly variable datasets. Current challenges include developing algorithms that maintain high fault detection precision across diverse conditions. Moreover, advanced signal processing methods, such as wavelet transforms and Kalman filters, could be leveraged as effective preprocessing techniques to enhance data coherence, reduce noise artifacts, and facilitate more reliable fault diagnosis. Future research should focus on improving the robustness of ML/DL algorithms for real-world operational environments, designing lightweight models suitable for deployment on edge devices, and developing hybrid approaches that integrate advanced signal processing with machine learning to enhance system resilience and precision.

7.3 Gaps and future research directions

Future research in wayside condition monitoring should prioritize the development of more portable and flexible sensing technologies. In particular, enhancing the portability of FBG sensors could simplify installation procedures and minimize operational disruptions. Given the high reliability of FBG-acquired data, there is significant potential to develop new predictive methods for estimating the residual useful life of rolling stock components and assessing derailment risks.

Furthermore, integrating data from diverse sensor types—such as FBGs, vision sensors, and high-speed cameras—offers opportunities to achieve a more comprehensive and multi-dimensional understanding of railway conditions. Research into advanced algorithmic techniques for regions of interest (ROI) segmentation, data quality enhancement, and precise wheel profile estimation remains limited and requires further exploration. Simultaneously considering geometric wheel profile parameters alongside tread defects could provide richer and more actionable insights into wheel and rail interactions.

Recent advancements in deep learning have revolutionized pattern recognition, enabling the extraction of meaningful features even from noisy or complex datasets. However, DL applications in wayside monitoring are still in their early stages. Future work should aim to develop more robust DL models capable of detecting a wider range of faults under varying operational conditions. In addition, evolving techniques for extracting three-dimensional information from two-dimensional image data [259] could eventually eliminate the need for auxiliary laser equipment, reducing system

complexity and cost. Although some research has enabled the identification of wheel flats [9, 27, 30, 77–80, 84, 188, 199, 213, 259], polygonal wheels [8, 76], and general damage diagnosis [8, 76, 191], more powerful machine learning-based algorithms are still required for accurate localization, prediction, and comprehensive diagnosis of various defect types. Future advancements in Railway Wayside Condition Monitoring systems should consider the digitalization potential of unmanned aerial vehicles (UAVs), unmanned aerial systems (UASs), and portable smartphones to enhance monitoring capabilities. Autonomous damage detection methods using computer vision and deep learning techniques, such as graphics-based digital twins [260, 261], are also beneficial for developing modern advanced WCWs. These technologies can significantly improve data collection, offering real-time, high-resolution insights into the condition of railway infrastructure.

In addition, investment in advanced ML algorithms is essential to effectively process and interpret the complex datasets generated by these digital tools. As digital tools and wireless communications become integral to WCM systems, robust cybersecurity measures must be prioritized to protect sensitive operational data and prevent malicious disruptions. Robust cybersecurity measures must be implemented to protect sensitive data and maintain system integrity. Embracing the mobility and flexibility of UAVs, UASs, and smartphones can lead to more comprehensive, efficient, and cost-effective monitoring solutions, ultimately enhancing the safety and reliability of railway networks. To provide a structured overview, Table 10 summarizes the main gaps identified in current wayside condition monitoring systems, outlines corresponding future research directions,

and suggests potential technical solutions to address these challenges.

8 Conclusion

This paper has presented a comprehensive review of wayside condition monitoring systems for railway vehicles. A detailed bibliometric analysis highlights the growing research interest in this field, reflecting the increasing importance of intelligent monitoring solutions to enhance railway safety, operational efficiency, and maintenance practices. The review systematically examined current commercial WCM systems, their sensor technologies, and the machine learning algorithms employed for data analysis and fault diagnosis. Special attention was given to common rolling stock defects—including wheel flats, polygonization, shelling, bearing faults, hunting instability, and load-related issues—and how WCM systems detect these anomalies.

The key conclusion drawn from this work is that while WCM systems have achieved significant maturity, several challenges persist. These include limitations in early stage defect detection, real-time processing, generalization of data across diverse operational conditions, and seamless integration into railway infrastructure. The significance of these findings lies in guiding future research towards developing more robust, scalable, and predictive monitoring solutions that can adapt to the increasing demands of modern railways.

Nonetheless, the findings presented are subject to certain limitations. The performance of WCM technologies heavily depends on environmental conditions, sensor reliability, data availability, and the quality of ML models. Moreover, many

Table 10 Research roadmap summarizing key gaps, future directions, and potential technical solutions for advancing wayside condition monitoring systems

Identified gap	Future research direction	Potential solutions and technologies
Limited portability and installation complexity of FBG sensors	Develop portable, easily deployable FBG sensor systems	Wireless FBG sensors, modular sensing units, flexible installation kits
Lack of integration between multiple sensor types (FBG, vision, cameras)	Design multi-sensor fusion frameworks for comprehensive condition monitoring	Sensor data fusion algorithms, multimodal AI architectures
Limited methods for simultaneous analysis of wheel profile and tread faults	Develop integrated analysis techniques combining geometry and surface defects	Hybrid machine learning models, multi-task learning frameworks
Immaturity of deep learning applications for wayside monitoring	Advance robust DL architectures capable of handling noisy and diverse datasets	Data augmentation, domain adaptation, semi-supervised DL approaches
Challenges in 3D information extraction from 2D vision data	Improve 3D reconstruction techniques without relying on external laser systems	Monocular depth estimation, 2D-to-3D AI modeling
Insufficient diagnostic capabilities for complex damage patterns (e.g., flats, polygonal wheels, ...)	Enhance ML models for comprehensive fault localization, prediction, and diagnosis	Ensemble learning, graph-based neural networks, temporal sequence modeling
Underutilization of UAVs, UASs, and mobile devices for monitoring	Using mobile platforms for flexible, real-time infrastructure inspection	UAV-based CV systems, smartphone-integrated monitoring apps
Growing cybersecurity risks in WCM systems	Implement robust cybersecurity measures to protect data integrity and system operations	Blockchain for data integrity, secure communication protocols, AI-driven intrusion detection

of the advanced methodologies reviewed are still primarily validated on synthetic or controlled datasets rather than in large-scale field deployments. Future research should focus on addressing these challenges through interdisciplinary efforts that combine railway engineering, data science, and real-world system validation. Particularly, there is a need for standardized benchmarks, collaborative data sharing, and the development of resilient algorithms capable of coping with the complex dynamic environment of railway operations.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose. The authors also confirm that Prof. Zhai, the Editor-in-Chief of Railway Engineering Science, was excluded from the editorial review or decision process for this manuscript.

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